

Stable Self-Similar Expansion of Four and Five Point Vortices and Confinement of Vortex Patches

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Abstract

Point vortices can move self-similarly, keeping their shape while the configuration grows like $\sqrt{t}$ and turns. For three vortices such expanding configurations are stable, and Zbarsky used this to show that vortex patches placed at the three vortices stay confined for all time; he wrote that the result would most likely carry over to stably growing systems of four or more vortices, given sufficiently good stability. We give such configurations of four and five vortices, with the circulations $(-1, -5/2, -1/9, 4/5)$ and $(-1, 3/7, 7/8, -9/7, -47/35)$. In similarity variables their linearization has, besides the double eigenvalue 0 of the rotation and of the one-parameter family of such configurations, only eigenvalues with real part -1 or -2 . The proof is computer-assisted: the configurations are enclosed by the Krawczyk method in ball arithmetic, and the spectrum is controlled through an exact lemma, which shows that six eigenvalues are forced by the rotation, the scaling, the translations and the family and that the others come in pairs $k, 2 - k$, together with enclosures of traces of powers of the Jacobian. The stability is nonlinear: every motion with the same circulations that starts near one of the two configurations stays within a bounded distance of an exactly self-similar expansion of a nearby member of the family, the member with the same energy, and an a priori estimate of the same kind holds for approximate solutions whose error is small and decays faster than the velocities. With this estimate in place of the one step of Zbarsky's proof that needs three vortices, his confinement theorem holds for the two configurations: vortex patches placed at the vortices stay within distance $\varepsilon t^{1/4+\varepsilon}$ of their centres of vorticity for all time, and the centres stay within a bounded distance of an exactly self-similar expansion of a single member of the family. The proof carries his argument over to any number of patches, with every estimate written out. Linear stability is an open condition; in a naive random search, 52 of 342 converged four-vortex collapses and 32 of 543 converged five-vortex collapses, not checked for duplicates, reverse into linearly stable expansions.

Keywords: point vortices; self-similar motion; nonlinear stability; vortex patches; computer-assisted proof; interval arithmetic.

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1 Introduction

N point vortices with circulations Γ_j at positions $z_j \in \mathbb{C}$ move by

$$\overline{\dot{z}_j} = \frac{1}{2\pi i} \sum_{l \neq j} \frac{\Gamma_l}{z_j - z_l}. \quad (1)$$

A motion is self-similar when the configuration keeps its shape while it grows or shrinks and turns; this happens when $\dot{z}_j = \kappa(z_j - z_c)$ for all j at $t = 0$, with one $\kappa \in \mathbb{C}$, where $z_c = \sum_j \Gamma_j z_j / \sum_j \Gamma_j$ [8, Sect. 2]. Then the size of the configuration grows or shrinks like $|1 + 2t \operatorname{Re} \kappa|^{1/2}$, the configuration expands when $\operatorname{Re} \kappa > 0$ and collapses in finite time when $\operatorname{Re} \kappa < 0$ [11, 14, 2, 6, 10]. Reversing every circulation reverses time, so each collapse becomes an expansion of the same shape, and conversely.

Zbarsky [17] proved that for a generic self-similarly expanding configuration of three vortices one may replace each vortex by a small patch of vorticity and the patches stay confined for all time, their radius growing more slowly than their distance. The restriction to three vortices comes from the stability analysis. He writes that “if one found stably growing systems of 4 or more vortices, the confinement result would most likely carry over with little modification. However, one needs sufficiently good stability results; orbital linear stability, which may not be hard to obtain for some systems, is not enough” [17, pp. 4–5 of the arXiv version]; the same point recurs in [18, after Cor. 16]. For three vortices, stability of the expanding family goes back to Tavantzis and Ting [15]; Iwayama and Yajima studied the linear stability of the self-similar motions of three vortices in the α -models [9]. Leoncini, El Kettani and Ugalde [12] split a vortex into three or four vortices that expand from a point by reversing a collapse, inside a larger system of vortices, and observe numerically, with a first-order argument for times of order the inverse of the relative size of the new vortices, that the other vortices do not feel the split; they do not study the stability of the expanding shape.

We show that stable expansions of four and five vortices exist, and that Zbarsky’s confinement theorem holds for them. Section 2 sets up similarity variables, in which a self-similar motion is a fixed point, and proves a lemma on the structure of the spectrum of the linearization (Lemma 1): six eigenvalues are forced by the rotation, the scaling, the translations and the one-parameter family of self-similar configurations with the given circulations (the second eigenvalue 2 through a pairing), and the others come in pairs $k, 2 - k$. For three vortices nothing else is left. For four vortices one pair is left, and a single number decides its stability; for five, two pairs. Section 3 states and proves, with a computer-assisted proof, that there are configurations of four and of five vortices whose reversal expands and whose exponents other than the six forced ones all have real part 1 (Theorems 1 and 2, Corollary 1). Section 4 proves that the stability is nonlinear, with the sharp rate and a limit fixed by the energy (Theorem 3, Corollary 2), together with an a priori estimate for approximate solutions (Proposition 1). Section 5 uses that estimate in place of the one step of Zbarsky’s proof that needs three vortices and obtains confinement of vortex patches for the two configurations (Theorem 4). Section 6 gives numerical illustrations, controls and a sample of how common such configurations are, and Section 7 discusses what is and is not shown.

2 Similarity variables and the structure of the spectrum

2.1 Similarity variables

Let $V(\zeta) = (V_1, \dots, V_N)$ be the velocity given by (1), $V_j = (i/2\pi) \sum_{l \neq j} \Gamma_l / (\zeta_j - \zeta_l)$. It is homogeneous of degree -1 , $V(\lambda\zeta) = \lambda^{-1}V(\zeta)$ for real $\lambda > 0$, and equivariant under rotations, $V(e^{i\phi}\zeta) = e^{i\phi}V(\zeta)$. For $\kappa = a + i\Omega$ put $z_j(t) = \rho(t)e^{i\Omega\tau}\zeta_j(\tau)$ with $\rho^2 = 1 + 2at$ and $d\tau/dt = \rho^{-2}$; then (1) becomes

$$\frac{d\zeta}{d\tau} = V(\zeta) - \kappa\zeta, \quad (2)$$

and a self-similar motion with this κ and $z_c = 0$ is a fixed point of (2). For a collapse we normalize $2\pi a = -1$; since κ scales like the inverse square of length, this fixes, given the circulations, the length scale of the configuration. We write $2\pi\kappa = -1 + ib$, and work with $E(\zeta) = 2\pi(V(\zeta) - \kappa\zeta) =$

$2\pi V(\zeta) + (1 - ib)\zeta$ and the rescaled time $s = \tau/2\pi$; for a collapse, $s = -\ln \rho$. The winding number of the collapse is $P = |b|/2$ [8]. Let DE be the real $2N \times 2N$ Jacobian of E at a zero ζ^* . A perturbation along an eigenvector with eigenvalue k behaves like $e^{ks} = \rho^{-k}$ relative to the size ρ of the configuration. In the real coordinates $(\text{Re } \delta\zeta, \text{Im } \delta\zeta)$ one finds

$$DE = \begin{pmatrix} A_r + I & A_i + bI \\ A_i - bI & -A_r + I \end{pmatrix}, \quad A_{jl} = \frac{i\Gamma_l}{(\zeta_j - \zeta_l)^2} \ (l \neq j), \quad A_{jj} = -\sum_{l \neq j} A_{jl}, \quad (3)$$

with $A = A_r + iA_i$ evaluated at ζ^* .

Time reversal. If ζ^* is a collapse for the circulations Γ , it is a fixed point of the similarity dynamics of the circulations $-\Gamma$ with κ replaced by $-\kappa$, an expansion with the same winding, and the Jacobian of its similarity dynamics is $-DE$; the similarity time is then $s = \ln \rho$. So a mode with eigenvalue k of DE behaves in the expansion like ρ^{-k} relative to its size, and it decays as the configuration grows exactly when $\text{Re } k > 0$.

2.2 The structure of the spectrum

Lemma 1. *Let ζ^* be a zero of E , with all $\Gamma_j \neq 0$ and the ζ_j^* distinct, and suppose that near (ζ^*, b) the zeros (ζ, b) of E , with the circulations fixed, form up to rotation a smooth curve through (ζ^*, b) whose ζ -component is transverse to the rotations. Then the characteristic polynomial p of DE satisfies $p(k) = p(2 - k)$ and is divisible by*

$$k^2(k - 2)^2((k - 1)^2 + b^2),$$

and the quotient q , of degree $2N - 6$, also satisfies $q(k) = q(2 - k)$. In particular the $2N - 6$ remaining eigenvalues, counted with multiplicity, come in pairs $k, 2 - k$.

Proof. Rotation. $E(e^{i\phi}\zeta) = e^{i\phi}E(\zeta)$, so differentiating at $\phi = 0$ gives $DE(i\zeta^*) = iE(\zeta^*) = 0$.

Scaling. $E(\lambda\zeta) = 2\pi V(\zeta)/\lambda + (1 - ib)\lambda\zeta$; at $\lambda = 1$ the derivative is $-2\pi V(\zeta^*) + (1 - ib)\zeta^* = 2(1 - ib)\zeta^*$, since $2\pi V(\zeta^*) = -(1 - ib)\zeta^*$. So $DE \zeta^* = 2\zeta^* - 2bi\zeta^*$, and on the span of ζ^* and $i\zeta^*$ the map DE is triangular with eigenvalues 2 and 0.

Translations. V is invariant under a common translation of all ζ_j , so DE acts on $\mathbf{1} = (1, \dots, 1)$ and $i\mathbf{1}$ as multiplication by $1 - ib$: $DE \mathbf{1} = \mathbf{1} - b i \mathbf{1}$, $DE(i\mathbf{1}) = i\mathbf{1} + b \mathbf{1}$, with eigenvalues $1 \pm ib$.

The family. Along the curve of zeros, $E(\xi(\sigma); b(\sigma)) = 0$, and $\partial_b E = -i\zeta$. Differentiating, $DE \xi'(0) = b'(0) i\zeta^*$, so $DE^2 \xi'(0) = 0$. As $\xi'(0)$ is not a real multiple of $i\zeta^*$, the generalized kernel of DE has dimension at least 2: the algebraic multiplicity of 0 is at least 2.

Pairing. With $H = -\frac{1}{2} \sum_{j \neq l} \Gamma_j \Gamma_l \ln |\zeta_j - \zeta_l|$, the field $\zeta \mapsto 2\pi V(\zeta) - ib\zeta$ is the Hamiltonian vector field of $H + (b/2) \sum_j \Gamma_j |\zeta_j|^2$ for the symplectic form $\sum_j \Gamma_j dx_j \wedge dy_j$, which is nondegenerate because every $\Gamma_j \neq 0$ (Kirchhoff's form of (1) and the rotations generated by the angular impulse $\sum_j \Gamma_j |\zeta_j|^2$, up to the sign convention of the form). The linearization L of a Hamiltonian vector field at any point is an infinitesimally symplectic matrix: JL is symmetric for the matrix J of the form, so $-L^\top = J L J^{-1}$, and L is similar to $-L$, Jordan blocks included; in particular its characteristic polynomial is even. The linearization of this field at ζ^* is $L = DE - I$, so $p(k) = r(k - 1)$ with r even, that is $p(k) = p(2 - k)$, with multiplicities, and DE is similar to $2I - DE$.

Conclusion. By pairing, the multiplicity of 2 equals that of 0, at least 2, so $k^2(k - 2)^2$ divides p . The span of $\mathbf{1}$ and $i\mathbf{1}$ is invariant, and the characteristic polynomial of DE on it, $(k - 1)^2 + b^2$, divides p too; its roots $1 \pm ib$ are never 0 or 2, so the two factors are coprime and p is divisible by the displayed product. That product is invariant under $k \mapsto 2 - k$, hence so is the quotient. $\square$

For three vortices $2N - 6 = 0$: in similarity variables every exponent of a three-vortex expansion is forced by the symmetries and the family, which is the linear shadow of the classical stability of the expanding three-vortex family [15]. For four vortices one pair k , $2 - k$ remains, and with $c = k(2 - k)$ the sum of squares of all eigenvalues gives

$$\text{tr}(DE^2) = (0 + 0 + 4 + 4 + 2(1 - b^2)) + (4 - 2c), \quad c = 3 + 3b^2 - \text{tr}(A\bar{A}), \quad (4)$$

using $\text{tr}(DE^2) = 2\text{tr}(A\bar{A}) + 2N(1 - b^2)$, which follows from (3). Both remaining exponents have positive real part exactly when $c > 0$; when $c > 1$ they are $1 \pm i\sqrt{c - 1}$. For five vortices two pairs remain, and the values of $u = (k - 1)^2$ on them are the roots u_1, u_2 of $u^2 - m_1u + m_2$. The six forced eigenvalues contribute $4 - 2b^2$ to $\text{tr}((DE - I)^2)$ and $4 + 2b^4$ to $\text{tr}((DE - I)^4)$, and a remaining pair contributes $2u$ and $2u^2$, so

$$m_1 = u_1 + u_2 = \frac{1}{2}(\text{tr}((DE - I)^2) - 4 + 2b^2), \quad m_2 = u_1u_2 = \frac{1}{2}(m_1^2 - p_2), \quad (5)$$

with $p_2 = u_1^2 + u_2^2 = \frac{1}{2}(\text{tr}((DE - I)^4) - 4 - 2b^4)$. Every remaining exponent has positive real part exactly when every root u satisfies $|\text{Re } \sqrt{u}| < 1$; in particular it suffices that both roots are real and less than 1.

3 Results

Theorem 1 (Four vortices). *For the circulations $\Gamma = (1, 5/2, 1/9, -4/5)$ there is a self-similar collapse of four Euler point vortices with $2\pi\kappa = -1 + ib$, winding*

$$P = b/2 = 2.5258464741344909107270\dots,$$

and z_4 with real part $8/25$ in the gauge $z_1 \in \mathbb{R}$, $z_c = 0$, whose Jacobian DE has the six eigenvalues $0, 0, 2, 2, 1 \pm ib$ of Lemma 1 and the pair $1 \pm i\vartheta$ with $\vartheta = 4.4588620962481\dots$; here $c = 20.881451193358\dots$ in (4).

Theorem 2 (Five vortices). *For the circulations $\Gamma = (1, -3/7, -7/8, 9/7, 47/35)$ there is a self-similar collapse of five Euler point vortices with winding $P = 2.40234703541713841\dots$ and z_2 with real part $3/5$ in the same gauge, whose Jacobian DE has the six eigenvalues of Lemma 1 and the four eigenvalues $1 \pm 3.4538555\dots i$ and $1 \pm 3.1885675\dots i$.*

Corollary 1. *For the circulations $(-1, -5/2, -1/9, 4/5)$ and $(-1, 3/7, 7/8, -9/7, -47/35)$ the configurations of Theorems 1 and 2 expand self-similarly, growing like $\sqrt{t}$. In similarity variables the linearization at them has the double eigenvalue 0, whose generalized eigenspace consists of the rotation and the tangent of the one-parameter family of expanding configurations with the same circulations, and every other eigenvalue has real part -1 or -2 . In the invariant complement of the rotation and the family, every solution of the linearized equation therefore decays, relative to the size ρ of the configuration, at least like ρ^{-1} , that is like $t^{-1/2}$.*

Proof of the theorems. The program `verify_stable_expansion.py` (Section 6) works in the gauge of [8, Sect. 8]: $\Gamma_1 = 1$, z_1 real and

$$\sum_{l \neq j} \frac{\Gamma_l}{z_j - z_l} = (2P - i) \bar{z}_j \quad (j = 1, \dots, N), \quad (6)$$

which is $E = 0$ with $b = 2P$ and $z_c = 0$. In the gauge P is signed, $b = 2P$, and $|P|$ is the winding number of Section 2. Three necessary conditions hold exactly at every zero of (6). Multiplying (6)

by Γ_j and summing over j gives $0 = (2P - i) \sum_j \Gamma_j \bar{z}_j$, because the left side is antisymmetric in j and l , so $\sum_j \Gamma_j z_j = 0$. Multiplying by $\Gamma_j z_j$ and summing gives $\sum_{j < l} \Gamma_j \Gamma_l = (2P - i) \sum_j \Gamma_j |z_j|^2$, because $z_j/(z_j - z_l) + z_l/(z_l - z_j) = 1$; both sums are real and $2P - i$ has imaginary part -1 , so $\sum_j \Gamma_j |z_j|^2 = 0$, and then $\sum_{j < l} \Gamma_j \Gamma_l = 0$.

For four vortices the program fixes $x_4 = 8/25$, $\Gamma_2 = 5/2$ and $\Gamma_3 = 1/9$, so that (6) becomes 8 real equations in the 8 unknowns $x_1, x_2, y_2, x_3, y_3, y_4, \Gamma_4$ and P , where $z_j = x_j + iy_j$. It proves by the Krawczyk test, in the ball arithmetic of FLINT/Arb, that these equations have exactly one zero in the box of radius 10^{-4} in the maximum norm about the point computed by Newton's method, and that every Jacobian of the system on this box is invertible. The tightened Krawczyk enclosure contains the zero and has radius below 10^{-93} ; the side conditions and every enclosure below are evaluated on it. On it the circulations are nonzero, the total circulation is nonzero, the vortices are distinct and away from the origin, $P > 0$, the three necessary conditions are enclosed with 0, as they must be, and for each j the enclosure of the quotient of the left side of (6) by $\bar{z}_j$ overlaps that of $2P - i$; these are the side conditions. With Γ_1, Γ_2 and Γ_3 fixed, $\sum_{j < l} \Gamma_j \Gamma_l = 26/9 + (65/18)\Gamma_4$ is linear in Γ_4 , so $\Gamma_4 = -4/5$ at the zero, and the enclosure of Γ_4 , which contains $-4/5$, is a consistency check. By the implicit function theorem the zeros with Γ_2 and Γ_3 fixed form a smooth curve parametrized by x_4 ; along it $\Gamma_4 = -4/5$ for the same reason, and its tangent v has $\text{Im } v_1 = 0$ and v_4 with real part 1, while $i\zeta^*$ has $\text{Im}(i\zeta_1^*) = x_1 \neq 0$, so v is not a real multiple of $i\zeta^*$. The hypotheses of Lemma 1 hold, and the program also checks on the enclosure that the vectors of its proof satisfy their identities; one of them is $DEv = b'(0)i\zeta^*$, where $b'(0)$ is the derivative of b along the curve with respect to x_4 , and the program encloses $b'(0) = 0.94234865\dots$ away from 0. It then encloses $c = 3 + 3b^2 - \text{tr}(A\bar{A})$, printed to 15 digits as $20.8814511933587 \pm 1.4 \cdot 10^{-14}$ (the enclosure itself is far tighter), so $c > 1$ and the remaining pair is $1 \pm i\sqrt{c-1}$. It also evaluates $(\text{tr}((DE - I)^2) - 4 + 2b^2)/2$ and finds $1 - c$; since $\text{tr}((DE - I)^2) = 2\text{tr}(A\bar{A}) - 2Nb^2$ for every configuration, this is the same number, and the agreement is a regression test of how the program assembles DE , not independent evidence.

For five vortices the program fixes $x_2 = 3/5$, $\Gamma_2 = -3/7$, $\Gamma_3 = -7/8$ and $\Gamma_4 = 9/7$ and proves in the same way that the 10 equations in the 10 unknowns $x_1, y_2, x_3, y_3, x_4, y_4, x_5, y_5, \Gamma_5$ and P have exactly one zero in the box of radius 10^{-5} in the maximum norm about the Newton point; the tightened enclosure again has radius below 10^{-93} . Here $\sum_{j < l} \Gamma_j \Gamma_l = -517/392 + (55/56)\Gamma_5$, so $\Gamma_5 = 47/35$ at the zero, and the program checks the side conditions and the hypotheses of the lemma as before and encloses $b'(0) = -9.6795840\dots$, the derivative with respect to x_2 , away from 0. It encloses m_1 and m_2 of (5); the discriminant $m_1^2 - 4m_2$ is positive and both roots, $u_1 = -11.929118\dots$ and $u_2 = -10.166963\dots$, are negative, so the four remaining eigenvalues are $1 \pm i\sqrt{-u_1}$ and $1 \pm i\sqrt{-u_2}$. $\square$

Proof of the corollary. By the time reversal of Section 2 the configurations expand for the negated circulations, with Jacobian $-DE$ in similarity variables and similarity time $s = \ln \rho$. The eigenvalues of $-DE$ are 0, 0, -2 , -2 , $-1 \mp ib$ and, by the theorems, $-1 \mp i\vartheta$ or $-1 \mp i\sqrt{-u_{1,2}}$. By the theorems no remaining eigenvalue is 0, so the generalized eigenspace of 0 is exactly two-dimensional and, by the proof of Lemma 1, spanned by $i\zeta^*$ and the tangent of the family. The eigenvalues with real part -1 are simple: $\vartheta \neq b$ for four vortices and $\sqrt{-u_1}, \sqrt{-u_2}$ and b are distinct for five, which the program checks. The eigenvalue -2 is double, the partner of the one at 0, and like it a single Jordan block of size 2: the tangent v of the family satisfies $DEv = b'(0)i\zeta^*$ with $b'(0) \neq 0$, which the program encloses away from 0 (proof of the theorems), so the kernel of DE is spanned by $i\zeta^*$ alone, and DE is similar to $2I - DE$ (proof of Lemma 1), which carries the Jordan blocks of DE at 0 to those at 2. It contributes se^{-2s} . On the complementary invariant subspace every solution of the linearized equation therefore decays at least like $e^{-s} = \rho^{-1}$, and ρ^2 grows linearly in t . $\square$

Remark 1 (Open sets). The conclusions are open conditions: c , and the roots u , depend continuously on the configuration and the circulations along the collapse set, which near these points is a smooth manifold of dimension $N - 1$ (modulo translations, rotations and the scalings of lengths and circulations) [8, Sect. 7]. So every collapse near those of the theorems, with nearby circulations, reverses into a linearly stable expansion as well; the stable expansions of four and five vortices form open subsets of the $(N - 1)$ -dimensional collapse set, not isolated examples.

4 Nonlinear stability

Throughout this section ζ^* is the configuration of Theorem 1 or of Theorem 2, b^* is its b , and the circulations are negated, so that ζ^* expands. By the time reversal of Section 2, a motion $z(t)$ of the vortices with circulations $-\Gamma$ is $z(t) = \rho(t)e^{-ib^*s}\zeta(s)$ with $\rho^2 = 1 + t/\pi$, $s = \ln \rho$ and

$$\frac{d\zeta}{ds} = -E(\zeta), \quad E(\zeta) = 2\pi V(\zeta) + (1 - ib^*)\zeta, \quad (7)$$

where V and E are formed with the circulations Γ of the theorems; this holds for every motion, not only for self-similar ones. We use the Euclidean norm $|\cdot|$ on $\mathbb{C}^N = \mathbb{R}^{2N}$, write $\mathcal{O} = \{e^{i\phi}\zeta^* : \phi \in \mathbb{R}\}$ for the orbit of ζ^* under rotations, and $H = -\sum_{j < l} \Gamma_j \Gamma_l \ln |\zeta_j - \zeta_l|$ for the energy, which is the same for Γ and $-\Gamma$. By the proof of the theorems, the zeros of $E(\zeta; b) = 2\pi V(\zeta) + (1 - ib)\zeta$ near (ζ^*, b^*) with z_1 real form a real-analytic curve $\sigma \mapsto (\xi(\sigma), b(\sigma))$, $|\sigma| < \sigma_1$, with $\xi(0) = \zeta^*$ and $b(0) = b^*$, parametrized by $\sigma = x_4 - 8/25$ or $\sigma = x_2 - 3/5$. Since $E(\zeta; b^*) = E(\zeta; b) + i(b - b^*)\zeta$, the solution of (7) starting at $\xi(\sigma)$ is $e^{-i(b(\sigma) - b^*)s}\xi(\sigma)$: another self-similar expansion, with a different shape and winding. We write ξ' for the derivative in σ and ζ' for the derivative of a solution in s . We fix $\sigma_0 \in (0, \sigma_1)$ once, small enough for Lemma 2 below and such that $H(\xi(\sigma))$ is strictly monotone for $|\sigma| \leq \sigma_0$; the latter is possible because the program encloses $dH(\xi(\sigma))/d\sigma$ at $\sigma = 0$ away from 0 ($-0.5133915789\dots$ for four vortices and $0.2322290983\dots$ for five).

Theorem 3 (Nonlinear stability). *There are $\delta_0 > 0$ and C such that every solution of (7) with $\delta := \text{dist}(\zeta(0), \mathcal{O}) \leq \delta_0$ exists for all $s \geq 0$, and there are σ_∞ with $|\sigma_\infty| \leq C\delta$ and $\phi_\infty \in \mathbb{R}$ with*

$$|\zeta(s) - e^{i(\phi_\infty - (b(\sigma_\infty) - b^*)s)}\xi(\sigma_\infty)| \leq C\delta e^{-s} \quad (s \geq 0).$$

Moreover σ_∞ is the unique σ with $|\sigma| \leq \sigma_0$ and $H(\xi(\sigma)) = H(\zeta(0))$.

The constants δ_0 and C are not explicit.

Corollary 2 (In the original variables). *Let $z(t)$ be a motion of N point vortices with the circulations $-\Gamma$ and $|z(0) - e^{i\phi}\zeta^*| \leq \delta \leq \delta_0$ for some ϕ . Then there are a member $\xi(\sigma_\infty)$ of the family, with the energy $H(z(0))$, and $\phi_\infty \in \mathbb{R}$ such that the exactly self-similar expansion $Z(t) = \rho(t)e^{i(\phi_\infty - b(\sigma_\infty)\ln \rho(t))}\xi(\sigma_\infty)$ satisfies $|z(t) - Z(t)| \leq C\delta$ for all $t \geq 0$. The vortices stay within a bounded distance of an exactly self-similar expansion whose size grows like $\sqrt{t}$, and the relative deviation is $O(t^{-1/2})$.*

Proof. Here $\zeta(0) = z(0)$. Multiplying the estimate of Theorem 3 by $\rho e^{-ib^*s} = e^s e^{-ib^*s}$ gives $|z(t) - Z(t)| \leq C\delta$. Z is a motion because $\xi(\sigma_\infty)$ is a zero of $E(\cdot; b(\sigma_\infty))$, so a fixed point of the similarity dynamics with $b(\sigma_\infty)$ in place of b^* , and $H(\xi(\sigma_\infty)) = H(\zeta(0))$ by the theorem. $\square$

The theorem follows from an a priori estimate for approximate solutions, whose constants do not depend on how long the approximate solution is followed; that is the form a continuation argument needs (Section 7).

Proposition 1 (Forced version). *For $\mu > 0$ and $0 < \nu < \min(\mu, 1)$ there are $\delta_1 > 0$ and C_1 , depending only on ζ^* , μ and ν , with the following property. Let $0 < S \leq \infty$ and let $\zeta : [0, S) \rightarrow \mathbb{C}^N$ be continuously differentiable with*

$$|\zeta'(s) + E(\zeta(s))| \leq \alpha e^{-\mu s} \quad (0 \leq s < S), \quad \delta := \text{dist}(\zeta(0), \mathcal{O}), \quad \delta + \alpha \leq \delta_1.$$

Then for $0 \leq s < S$ we have $\text{dist}(\zeta(s), \mathcal{O}) \leq C_1(\delta + \alpha)$, and the distance from $\zeta(s)$ to $\{e^{i\phi}\xi(\sigma) : \phi \in \mathbb{R}, |\sigma| \leq \sigma_0\}$ is at most $C_1(\delta + \alpha)e^{-\nu s}$. If $S = \infty$, there are σ_∞ with $|\sigma_\infty| \leq C_1(\delta + \alpha)$ and a continuous function ϕ with $|\zeta(s) - e^{i\phi(s)}\xi(\sigma_\infty)| \leq C_1(\delta + \alpha)e^{-\nu s}$. If $S = \infty$ and $\mu > 1$, there is moreover $\phi_\infty \in \mathbb{R}$ with

$$|\zeta(s) - e^{i(\phi_\infty - (b(\sigma_\infty) - b^*)s)}\xi(\sigma_\infty)| \leq C_1(\delta + \alpha)e^{-s} \quad (s \geq 0).$$

Suppose that z satisfies $\dot{z} = V_{-\Gamma}(z) + e(t)$ for $t \geq 0$, where $V_{-\Gamma}$ is the velocity (1) for the circulations $-\Gamma$ and e is a continuous forcing with $|e(t)| \leq K(1+t)^{-(1+\mu)/2}$. Then $\zeta(s) = \rho^{-1}e^{ib^*s}z(t)$ satisfies $\zeta' = -E(\zeta) + 2\pi\rho e^{ib^*s}e(t)$, and since $1+t \geq \rho^2$ the forcing is at most $2\pi K e^{-\mu s}$: the proposition applies with $\alpha = 2\pi K$ when $\delta + 2\pi K \leq \delta_1$. So the error must be small and decay faster than the velocities, which are of order $t^{-1/2}$.

That a manifold of equilibria whose transverse spectrum lies in the open left half-plane attracts every nearby solution exponentially, each to one of its equilibria, is classical [3]. We give the short proof because the family is a curve of equilibria only modulo the rotation, and because we need the forced version and the sharp rate.

Lemma 2 (Coordinates). *Let $W_0 = \text{span}_{\mathbb{R}}\{i\zeta^*, \xi'(0)\}$ and let W_1 be the sum of the generalized eigenspaces of DE at ζ^* for its nonzero eigenvalues, and fix a linear isomorphism $Q : \mathbb{R}^{2N-2} \rightarrow W_1$. For $\sigma_0, r_0 > 0$ small enough, $\Phi(\phi, \sigma, y) = e^{i\phi}(\xi(\sigma) + Qy)$ is a real-analytic diffeomorphism from $(\mathbb{R}/2\pi\mathbb{Z}) \times (-\sigma_0, \sigma_0) \times \{|y| < r_0\}$ onto an open neighborhood U of $\mathcal{O}$. In these coordinates a continuously differentiable $\zeta(s) = \Phi(\phi, \sigma, y) \in U$ satisfies*

$$(\phi', \sigma', y') = (h(\sigma, y), f(\sigma, y), g(\sigma, y)) + L_{\sigma, y}^{-1}(e^{-i\phi}(\zeta' + E(\zeta))), \quad (8)$$

where $L_{\sigma, y}(p_1, p_2, w) = p_1 i(\xi(\sigma) + Qy) + p_2 \xi'(\sigma) + Qw$ and $(h, f, g)(\sigma, y) = -L_{\sigma, y}^{-1}E(\xi(\sigma) + Qy)$ are real-analytic, with $h(\sigma, 0) = -(b(\sigma) - b^*)$, $f(\sigma, 0) = 0$ and $g(\sigma, 0) = 0$. The spectrum of $B(\sigma) = \partial_y g(\sigma, 0)$ consists of the eigenvalue -2 , of algebraic multiplicity 2, and of $2N - 4$ simple eigenvalues with real part -1 , and there is K_B with $\|e^{B(\sigma)s}\| \leq K_B e^{-s}$ for $s \geq 0$ and $|\sigma| \leq \sigma_0$.

Proof. By Corollary 1 the generalized kernel of DE is W_0 , so $\mathbb{C}^N = W_0 \oplus W_1$ with both summands invariant under DE . The differential of Φ at $(0, 0, 0)$ is $L_{0,0}$, which is invertible; by rotation equivariance the differential of Φ is invertible at every (ϕ, σ, y) with σ, y small. If $\Phi(\phi, \sigma, y) = \Phi(\tilde{\phi}, \tilde{\sigma}, \tilde{y})$ with small $\sigma, \tilde{\sigma}, y, \tilde{y}$, then $e^{i(\tilde{\phi}-\phi)}\zeta^*$ is close to ζ^* , so $\tilde{\phi} - \phi$ is close to $2\pi\mathbb{Z}$ because $\zeta^* \neq 0$, and the inverse function theorem at $(0, 0, 0)$ gives equality. This proves the first claim. Differentiating $\zeta = \Phi(\phi, \sigma, y)$ gives $\zeta' = e^{i\phi}L_{\sigma, y}(\phi', \sigma', y')$, and $E(\zeta) = e^{i\phi}E(\xi(\sigma) + Qy)$ by equivariance, which gives (8). At $y = 0$, $E(\xi(\sigma)) = i(b(\sigma) - b^*)\xi(\sigma) = L_{\sigma, 0}(b(\sigma) - b^*, 0, 0)$, which gives the values of h , f and g there.

For the spectrum, consider (8) without forcing, with b^* replaced by $b(\sigma)$ for a fixed σ . Rotation equivariance shows that this changes only the equation for ϕ , by an additive constant, and $\xi(\sigma)$ becomes a fixed point. Its linearization in the coordinates is block upper triangular with diagonal blocks $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $B(\sigma)$, because $f(\cdot, 0) = g(\cdot, 0) = 0$, and it is similar to $-DE(\xi(\sigma); b(\sigma))$. Lemma 1 applies at every point of the curve, so the spectrum of $B(\sigma)$ consists of -2 , -2 , $-1 \pm ib(\sigma)$ and the negatives of the remaining eigenvalues. At $\sigma = 0$ these have real part -1 and are simple, by

the theorems. Along the curve they stay there for small $|\sigma|$: for four vortices $c > 1$ is preserved, and for five a positive discriminant and two negative roots u are preserved, while being distinct is an open condition. The eigenvalue -2 keeps algebraic multiplicity 2 because the others stay away from it. So

$$e^{B(\sigma)s} = \sum_{j=1}^{2N-4} e^{-k_j(\sigma)s} \Pi_j(\sigma) + e^{-2s} (I + sM(\sigma)) \Pi_0(\sigma),$$

where $-k_1(\sigma), \dots, -k_{2N-4}(\sigma)$ are the simple eigenvalues of $B(\sigma)$, with $\operatorname{Re} k_j = 1$, the Π_j are their spectral projections, Π_0 is that of -2 and $M = (B + 2I)\Pi_0$ satisfies $M^2 = 0$. The projections are continuous in σ , so K_B exists after shrinking σ_0 . $\square$

Proof of Proposition 1. Choose $\gamma \in (\nu, 1)$ with $\gamma \neq \mu$. Every eigenvalue of $B(0)$ has real part at most -1 , so there is an inner product on $\mathbb{R}^{2N-2}$, with norm $|\cdot|_*$, in which $\langle B(0)y, y \rangle_* \leq -\frac{1+\gamma}{2}|y|_*^2$. Since $B(\sigma) \rightarrow B(0)$ and $g(\sigma, y) = B(\sigma)y + O(|y|^2)$, there are $\sigma_2 \in (0, \sigma_0]$ and $r_2 > 0$, depending on γ , such that $\langle g(\sigma, y), y \rangle_* \leq -\gamma|y|_*^2$, $|f(\sigma, y)| \leq C_f|y|_*$ and $\|L_{\sigma, y}^{-1}\| \leq C_L$ for $|\sigma| \leq \sigma_2$ and $|y|_* \leq r_2$, a part of the domain of Lemma 2. By the inverse function theorem and equivariance, $\operatorname{dist}(\zeta(0), \mathcal{O}) = \delta$ gives $\zeta(0) = \Phi(\phi_0, \sigma(0), y(0))$ with $|\sigma(0)| + |y(0)|_* \leq C_0\delta$ when δ is small.

Let $S_1 \leq S$ be the supremum of the s_1 such that $\zeta(s) \in \Phi(\mathbb{R} \times [-\sigma_2/2, \sigma_2/2] \times \{|y|_* \leq r_2/2\})$ for $0 \leq s \leq s_1$. On $[0, S_1]$ the forcing term in (8) has size at most $C_L\alpha e^{-\mu s}$ (in the norm $|\cdot|_*$, after enlarging C_L), so the upper right Dini derivative of $|y|_*$ satisfies $D^+|y|_* \leq -\gamma|y|_* + C_L\alpha e^{-\mu s}$; where $y = 0$ this holds because y' is then the forcing term alone. The comparison lemma gives

$$|y(s)|_* \leq e^{-\gamma s}|y(0)|_* + C_L\alpha \int_0^s e^{-\gamma(s-\tau)} e^{-\mu\tau} d\tau \leq C_0\delta e^{-\gamma s} + \frac{C_L\alpha}{|\gamma - \mu|} e^{-\min(\gamma, \mu)s} \leq C_2(\delta + \alpha)e^{-\nu s}.$$

Then $|\sigma'| \leq C_f|y|_* + C_L\alpha e^{-\mu s} \leq C_3(\delta + \alpha)e^{-\nu s}$, so $|\sigma(s) - \sigma(0)| \leq C_3(\delta + \alpha)/\nu$. If $\delta + \alpha$ is so small that $C_0\delta + C_3(\delta + \alpha)/\nu < \sigma_2/4$ and $C_0\delta + C_2(\delta + \alpha) < r_2/4$, then $\zeta(s)$ stays in the image of $\mathbb{R} \times [-\sigma_2/4, \sigma_2/4] \times \{|y|_* \leq r_2/4\}$ on $[0, S_1)$, a compact set inside the image of $\mathbb{R} \times (-\sigma_2/2, \sigma_2/2) \times \{|y|_* < r_2/2\}$, and by continuity $S_1 = S$. The two distance bounds follow, since $|\Phi(\phi, \sigma, y) - e^{i\phi}\zeta^*| \leq |\xi(\sigma) - \zeta^*| + |Qy|$ and $|\Phi(\phi, \sigma, y) - e^{i\phi}\xi(\sigma)| = |Qy|$. If $S = \infty$, $\sigma(s)$ converges to some σ_∞ with $|\sigma(s) - \sigma_\infty| \leq C_3(\delta + \alpha)e^{-\nu s}/\nu$, and $|\zeta(s) - e^{i\phi(s)}\xi(\sigma_\infty)| \leq |\xi(\sigma(s)) - \xi(\sigma_\infty)| + |Qy(s)|$.

For the sharp rate, let $\mu > 1$ and $S = \infty$; by what was shown with $\nu = 1/2$ (after decreasing δ_1 if necessary), $|\sigma(s) - \sigma_\infty| + |y(s)| \leq C(\delta + \alpha)e^{-s/2}$. Write $y' = B(\sigma_\infty)y + r(s) + F(s)$, where $r = (B(\sigma(s)) - B(\sigma_\infty))y + (g(\sigma, y) - B(\sigma)y)$ satisfies $|r(s)| \leq C(\delta + \alpha)e^{-s/2}|y(s)|$ and F , the forcing term of (8), satisfies $|F(s)| \leq C_L\alpha e^{-\mu s}$. By Lemma 2 and the variation of constants formula,

$$e^s|y(s)| \leq K_B|y(0)| + \frac{K_B C_L \alpha}{\mu - 1} + K_B C(\delta + \alpha) \int_0^s e^{-\tau/2} e^\tau |y(\tau)| d\tau,$$

and Gronwall's inequality gives $|y(s)| \leq C(\delta + \alpha)e^{-s}$. Then $|\sigma(s) - \sigma_\infty| \leq \int_s^\infty (C_f|y| + C_L\alpha e^{-\mu\tau}) d\tau \leq C(\delta + \alpha)e^{-s}$. With $\beta_\infty = b(\sigma_\infty) - b^*$, $\phi' = h(\sigma, y) + O(\alpha e^{-\mu s}) = -\beta_\infty + O((\delta + \alpha)e^{-s})$, so $\phi(s) + \beta_\infty s$ converges to some ϕ_∞ at the rate $(\delta + \alpha)e^{-s}$, and the estimate follows from $\zeta(s) = e^{i\phi(s)}(\xi(\sigma(s)) + Qy(s))$. $\square$

Proof of Theorem 3. Apply Proposition 1 with $\alpha = 0$, $\mu = 2$ and $\nu = 1/2$ to the solution on its maximal interval of existence. It stays in a compact set without collisions, so it exists for all $s \geq 0$, and the last part of the proposition gives the estimate, with $|\sigma_\infty| \leq C\delta$.

For the energy: the sum $\sum_{j < l} \Gamma_j \Gamma_l$ vanishes exactly for the circulations of the theorems, so $H(\lambda\zeta) = H(\zeta)$ for $\lambda > 0$; with the invariance of H under rotations and its conservation by (1), $H(\zeta(s))$ is constant along (7). As $e^{-i\phi(s)}\zeta(s) \rightarrow \xi(\sigma_\infty)$, $H(\xi(\sigma_\infty)) = H(\zeta(0))$. For δ_0 small,

$|\sigma_\infty| \leq C\delta_0 \leq \sigma_0$, and since σ_0 was chosen so that $H(\xi(\sigma))$ is strictly monotone on $|\sigma| \leq \sigma_0$, σ_∞ is determined by $H(\zeta(0))$. $\square$

Remark 2 (Why the energy varies along the family). Differentiating the identity $\nabla H(\zeta) \cdot E(\zeta) = 0$, which expresses the conservation of H and its invariance under rotations and scalings, at a zero of E shows that $\nabla H(\zeta^*)$ is a left null vector of DE ; it is not zero, since $2\pi V$ is, up to sign, the Hamiltonian vector field of H for the nondegenerate form $\sum_j \Gamma_j dx_j \wedge dy_j$ (proof of Lemma 1) and $2\pi V(\zeta^*) = -(1 - ib^*)\zeta^* \neq 0$. When $b'(0) \neq 0$, the kernel of DE is spanned by $i\zeta^*$ alone, and the range of DE is $W_1 \oplus \mathbb{R}i\zeta^*$, a hyperplane that does not contain $\xi'(0)$. As $\nabla H(\zeta^*)$ vanishes on that range, $dH/d\sigma = \nabla H(\zeta^*) \cdot \xi'(0) \neq 0$. The program encloses $b'(0)$ away from 0 as well ($0.94234865\dots$ and $-9.6795840\dots$; proof of the theorems) and checks that $\nabla H(\zeta^*)^\top DE$ contains 0. The limit of a perturbed expansion is therefore fixed by a conserved quantity, which Section 6 illustrates. The same left null vector gives the double eigenvalue 0 of Lemma 1 without the hypothesis on the family, at any zero of E with nonzero circulations and distinct ζ_j (there $\sum_{j < l} \Gamma_j \Gamma_l = 0$ by the proof of the theorems, so H is invariant under scalings): $\nabla H(\zeta^*) \cdot i\zeta^* = 0$ by the invariance under rotations, so if the kernel of DE is spanned by $i\zeta^*$ (otherwise there is nothing to prove), then $\nabla H(\zeta^*)$ spans the left kernel, $i\zeta^*$ lies in the range of DE , and 0 has algebraic multiplicity at least 2 (and hence, with the pairing, the whole conclusion of Lemma 1). We keep the hypothesis because Section 4 uses the family itself.

5 Vortex patches

We now replace the point vortices by patches of vorticity. The Euler equations $\partial_t \omega + u \cdot \nabla \omega = 0$, $u = K * \omega$, $K(x) = x^\perp / (2\pi|x|^2)$, move point vortices by (1), and have a unique global weak solution for bounded, compactly supported initial vorticity ([16]; for the whole plane [4, Theorems 1.6 and 3.3]); a solution $\omega(t) = \sum_j \omega_j(t)$ is the sum of the parts transported from the initial pieces ω_j . Let ζ^* , b^* and Γ be as in Section 4, and put $Y = \zeta^* / \sqrt{\pi}$. With the origin of time moved to the point of emergence, $\rho^2 = t/\pi$, the expansion of Corollary 2 is $x^*(t) = \sqrt{t} e^{-i(b^*/2)\ln(t/\pi)} Y$, a motion of point vortices with circulations $-\Gamma$. Let $\mathcal{V} = \{z \in \mathbb{C}^N : \sum_j \Gamma_j z_j = 0\}$, which contains Y and iY , and let $S_Y = \{z \in \mathcal{V} : \langle Y, z \rangle = |Y|^2\}$ with $\langle z, w \rangle = \sum_j \bar{z}_j w_j$, a real affine slice of dimension $2N - 4$ through Y , transverse to rotations and dilations. Every $x \in \mathcal{V}$ with $\langle Y, x \rangle \neq 0$ is $x = e^{i\beta} \gamma z$ with unique $\beta \in \mathbb{R}/2\pi\mathbb{Z}$, $\gamma > 0$ and $z \in S_Y$, namely

$$e^{i\beta} \gamma = \langle Y, x \rangle / |Y|^2, \quad z = |Y|^2 x / \langle Y, x \rangle, \quad (9)$$

and writing $x = \lambda e^{i\phi} (Y + w)$ with $\lambda > 0$ and $|w| \leq |Y|/2$ one finds

$$|z - Y| \leq 4|w|, \quad |\gamma/\lambda - 1| \leq |w|/|Y|. \quad (10)$$

Theorem 4 (Confinement of vortex patches). *Let ζ^* be the configuration of Theorem 1 or of Theorem 2, with Y and S_Y as above. Let $\varepsilon > 0$ be small enough, $M > 0$ and $r > 0$. There is T such that for every $t_0 > T$ the following holds. Let $\omega_1, \dots, \omega_N$ be bounded functions of definite sign with $\text{supp } \omega_j \subseteq B(\sqrt{t_0} Y_j, r)$, $\|\omega_j\|_{L^{2+\varepsilon}} \leq M$, $\int \omega_j = -\Gamma_j$ and $\sum_j \int x \omega_j(x) dx = 0$. Let $\omega(t) = \sum_j \omega_j(t)$, $t \geq t_0$, be the solution of the Euler equations with $\omega(t_0) = \sum_j \omega_j$, and let $\bar{x}_j(t) = -\Gamma_j^{-1} \int x \omega_j(t, x) dx$ be the centres of vorticity. Then for all $t \geq t_0$ and all j :*

- (i) $\omega_j(t)$ has the sign of $-\Gamma_j$ and $\text{supp } \omega_j(t) \subseteq B(\bar{x}_j(t), \varepsilon t^{1/4+\varepsilon})$;
- (ii) $\bar{x}(t) = e^{i\beta(t)} \gamma(t) z(t)$ with β, γ and $z(t) \in S_Y$ given by (9), and $|z(t) - Y| < \varepsilon$, $|\gamma(t)/\sqrt{t} - 1| < \varepsilon$;

(iii) there are σ_∞ with $|\sigma_\infty| < \varepsilon$, $\phi_\infty \in \mathbb{R}$ and a constant C such that the exactly self-similar expansion $Z(t) = \sqrt{t/\pi} e^{i(\phi_\infty - (b(\sigma_\infty)/2) \ln(t/\pi))} \xi(\sigma_\infty)$ of point vortices with circulations $-\Gamma$ satisfies $|\bar{x}(t) - Z(t)| \leq C$.

Parts (i) and (ii) are the conclusion of Zbarsky's theorem for three vortices [17, Theorem 2], with his generic expanding triangle replaced by the configurations of Theorems 1 and 2. Part (iii) adds that the centres of vorticity stay within a bounded distance of an exactly self-similar expansion of a single member of the family. Dávila, del Pino, Musso and Parmeshwar [5] construct solutions whose continuous vorticity stays for all time in balls of fixed radius 3ε around the points of an expanding three-vortex spiral, a stronger confinement than $t^{1/4+\varepsilon}$ and one that needs no stability; they remark that their construction applies to the four- and five-vortex spirals of Novikov and Sedov [14] under analogous conditions on the circulations. That is a construction of particular solutions; Theorem 4 is about every initial datum of the stated kind.

Proof. The proof is Zbarsky's bootstrap argument [17, Sects. 3–5] for N patches, and Appendix A proves the estimates it needs in full. Fix $\varepsilon \leq 1/18$ small enough, then an even integer n large enough depending on ε (Proposition 2), then χ small and T large. The bootstrap assumptions (B1)–(B5), stated at the start of Appendix A, hold at t_0 when T is large: (B1) by (10), since $|\bar{x}(t_0) - \sqrt{t_0}Y| \leq \sqrt{N}r$, and (B2), (B3) and (B5) because the supports lie within $2r$ of the centres. Suppose that one of them fails at a first time $T_* < \infty$. The quantities involved are continuous in t , so the assumptions hold on $[t_0, T_*)$, and at T_* with $\leq$ in place of $<$.

Where $N = 3$ entered. Zbarsky's proof uses three vortices in one place: to recover (B1), at the end of his Section 3. There he uses that the angular impulse and the energy of the centres are nearly conserved, and that for three vortices these two quantities, with the rotation and the dilation, are coordinates near the reference configuration [17, Lemma 1(ii)]; the dilation then follows from $d|\bar{x}_1|^2/dt$. For $N \geq 4$ the configurations modulo rotation and dilation have dimension $2N - 4 > 2$, and two nearly conserved quantities cannot pin them down. We use Proposition 1 instead. Two further facts enter, and both hold here: $\sum_{i < j} \Gamma_i \Gamma_j = 0$, exactly for our circulations (the program checks it in rational arithmetic; it is necessary for any self-similar expansion, since $H(\lambda\zeta) = H(\zeta) - \ln \lambda \sum_{i < j} \Gamma_i \Gamma_j$), which is used in Lemma 7; and $\sum_j \Gamma_j \neq 0$, which follows because $(\sum_j \Gamma_j)^2 = \sum_j \Gamma_j^2 + 2 \sum_{i < j} \Gamma_i \Gamma_j > 0$ (the sums are $253/90$ and $93/40$); it is what lets a translation put the centre of vorticity at the origin, $\sum_j \int x \omega_j = 0$, which the appendix assumes.

(a) *The centres move as point vortices up to a small error.* By Lemma 4, with V_i^0 the velocity of the i th vortex in (1) for the circulations $-\Gamma$ at the positions $\bar{x}$,

$$\frac{d\bar{x}}{dt} = V_{-\Gamma}(\bar{x}) + e(t), \quad |e(t)| \leq C_e t^{\varepsilon/2-3/2} \quad (t_0 \leq t \leq T_*),$$

with C_e depending on Y , ε , M and r but not on T_* .

(b) *Recovering (B1).* Put $\zeta(s) = \rho^{-1} e^{ib^*s} \bar{x}(t)$ with $\rho = \sqrt{t/\pi}$ and $s = \ln \rho$. The computation after Proposition 1 uses only $d\rho/dt = 1/(2\pi\rho)$, so $\zeta' = -E(\zeta) + 2\pi\rho e^{ib^*s} e(t)$, and $2\pi\rho|e(t)| \leq 2\pi C_e \rho(\pi\rho^2)^{\varepsilon/2-3/2} \leq 2\pi C_e e^{-(2-\varepsilon)s}$. In the shifted time $s - s_0$, $s_0 = \ln \rho(t_0)$, the hypothesis of Proposition 1 holds with $\mu = 2 - \varepsilon$, forcing $\alpha_1 = 2\pi C_e \rho(t_0)^{\varepsilon-2}$, and $\delta = \text{dist}(\zeta(s_0), \mathcal{O}) \leq |\bar{x}(t_0) - \sqrt{t_0}Y|/\rho(t_0) \leq \sqrt{\pi N} r t_0^{-1/2}$. Both tend to 0 as $t_0 \rightarrow \infty$, so for T large the proposition, applied with $\nu = 1/2$ on $[s_0, \ln \rho(T_*)]$ and extended to T_* by continuity, gives $\text{dist}(\zeta(s), \mathcal{O}) \leq C_1(\delta + \alpha_1)$ on $[t_0, T_*]$. Since $\min_\phi |\bar{x}(t) - \sqrt{t} e^{i\phi} Y| = \sqrt{t/\pi} \text{dist}(\zeta(s), \mathcal{O})$, and $\bar{x}(t) \in \mathcal{V}$ because $\sum_j \Gamma_j \bar{x}_j = -\sum_j \int x \omega_j$ is conserved and vanishes at t_0 , (10) with $\lambda = \sqrt{t}$ gives $|z - Y| \leq 4C_1(\delta + \alpha_1)/\sqrt{\pi}$ and $|\gamma/\sqrt{t} - 1| \leq C_1(\delta + \alpha_1)/(\sqrt{\pi}|Y|)$ on $[t_0, T_*]$. For T large both bounds are less than $\varepsilon^2/2$, so (B1) holds on $[t_0, T_*]$ with strict inequalities, in particular at T_* .

(c) *The other assumptions.* Propositions 2 (if $T_* \geq t_0 + t_0^{9/10}$) and 3 (otherwise) give (B2), (B3) and (B5) at T_* with strict inequalities, and (B4) holds because the vorticity is transported by a divergence-free field. This contradicts the choice of T_* , so every assumption holds for all $t \geq t_0$, which gives (i) and (ii).

(d) *Part (iii).* Now the estimate of (a) holds for all $t \geq t_0$, so Proposition 1 applies on $[s_0, \infty)$ with $\mu = 2 - \varepsilon > 1$, and its last part gives σ_∞ with $|\sigma_\infty| \leq C_1(\delta + \alpha_1) < \varepsilon$ and ϕ_1 with $|\zeta(s) - e^{i(\phi_1 - (b(\sigma_\infty) - b^*)(s - s_0))} \xi(\sigma_\infty)| \leq C_1(\delta + \alpha_1)e^{-(s - s_0)}$. Multiplying by ρe^{-ib^*s} , and using $\rho e^{-(s - s_0)} = \rho(t_0)$, gives $|\bar{x}(t) - Z(t)| \leq C_1(\delta + \alpha_1)\rho(t_0) \leq C_1(\sqrt{N}r + \alpha_1\rho(t_0))$, which is bounded, with $\phi_\infty = \phi_1 + (b(\sigma_\infty) - b^*)s_0$. Z is a motion by the proof of Corollary 2. $\square$

6 Numerical illustration, controls and a sample

Controls. The same program certifies three controls in the same way, with the same side conditions (for the third, $P \neq 0$ in place of $P > 0$). For each of them the Krawczyk test gives the family of the implicit function theorem, and z_1 is real and, by a side condition, nonzero, so the hypotheses of Lemma 1 hold as in the proof of the theorems. The first two are negative controls, collapses whose reversal is unstable, where unstable means linearly unstable: the linearization of the expansion in similarity variables has an eigenvalue with positive real part. A four-vortex collapse with $x_4 = 149/125$ and the circulations $(1, -1003/1000, 13/10, 10069/12970)$, the last forced by $\sum_{j < l} \Gamma_j \Gamma_l = 0$, has $c = -47.7346 \dots < 0$, so the test of the proof of Theorem 1 refuses it: its remaining pair is real, $k = -5.9810 \dots < 0$, and its reversal is unstable. A five-vortex collapse with $x_2 = -289/857$ and the circulations $(1, -133/849, 150/839, -61/876, 6868618/84905979)$, near a solution found by a random search, has $m_1^2 - 4m_2 < 0$ in (5), so the test in the proof of Theorem 2 refuses it at its first condition. The refusal is right: its roots are u and $\bar{u}$, not real, and the program encloses $\sqrt{u} = 5.02216 \dots + 4.28972 \dots i$, so $|\operatorname{Re} \sqrt{u}| > 1$; two of the remaining exponents $1 \pm \sqrt{u}$, $1 \pm \sqrt{\bar{u}}$ have real part $-4.02216 \dots$, and its reversal is unstable. A negative discriminant alone would not show this, since non-real roots with $|\operatorname{Re} \sqrt{u}| < 1$ give four exponents with positive real part, which is why $|\operatorname{Re} \sqrt{u}| > 1$ is enclosed. The third is a positive control, a consistency check that must pass: a three-vortex collapse with $x_2 = 9/10$ and $\Gamma_2 = -238/125$ has $\operatorname{tr}((DE - I)^2) = 4 - 2b^2$ to within its enclosure, the contribution of the six forced eigenvalues alone, as Lemma 1 requires for $N = 3$. It turns the other way, $P < 0$ in the gauge (6), which is why it has $P \neq 0$ in place of the side condition $P > 0$. The program runs two more controls. The recipe (5) applied to the matrix DE of Theorem 1 gives $m_1 = 1 - c$ and $m_2 = 0$, and applied to the three-vortex control $m_1 = p_2 = 0$, to within the enclosures, as Lemma 1 requires. And the Krawczyk test fails, as it must, on the box of radius 10^{-6} about a point at distance 10^{-3} in the maximum norm from the zero of Theorem 1, a box that does not contain that zero.

Direct integration (binary64, DOP853, relative tolerance 10^{-12} , and 10^{-13} for the longest run; not part of the proof). The four-vortex expansion of Corollary 1, started from the midpoint of the certified enclosure, keeps its shape to $4 \cdot 10^{-15}$ while its size grows 178-fold; a random perturbation of 10^{-5} in each coordinate (a shape deviation of $3.3 \cdot 10^{-5}$) stays below $2 \cdot 10^{-4}$, settling on a nearby member of the family, the neutral direction. Measured against the member with the same energy, found by least squares as Theorem 3 prescribes, a perturbation of 10^{-4} in each coordinate has a shape deviation (modulo translation, rotation and scaling, so weaker than the quantity in the theorem) that decays like the inverse of the size. Followed from 5-fold to $5 \cdot 10^8$ -fold growth, deviation times size stays between $3.0 \cdot 10^{-4}$ and $6.3 \cdot 10^{-4}$ for four vortices, and between $3.1 \cdot 10^{-3}$ and $2.1 \cdot 10^{-2}$ for five, where two oscillating modes of nearly equal frequency beat with a period of about 24 in $\ln \rho$. The deviation from ζ^* itself levels off, near $1.1 \cdot 10^{-3}$ and $5.8 \cdot 10^{-3}$, and H

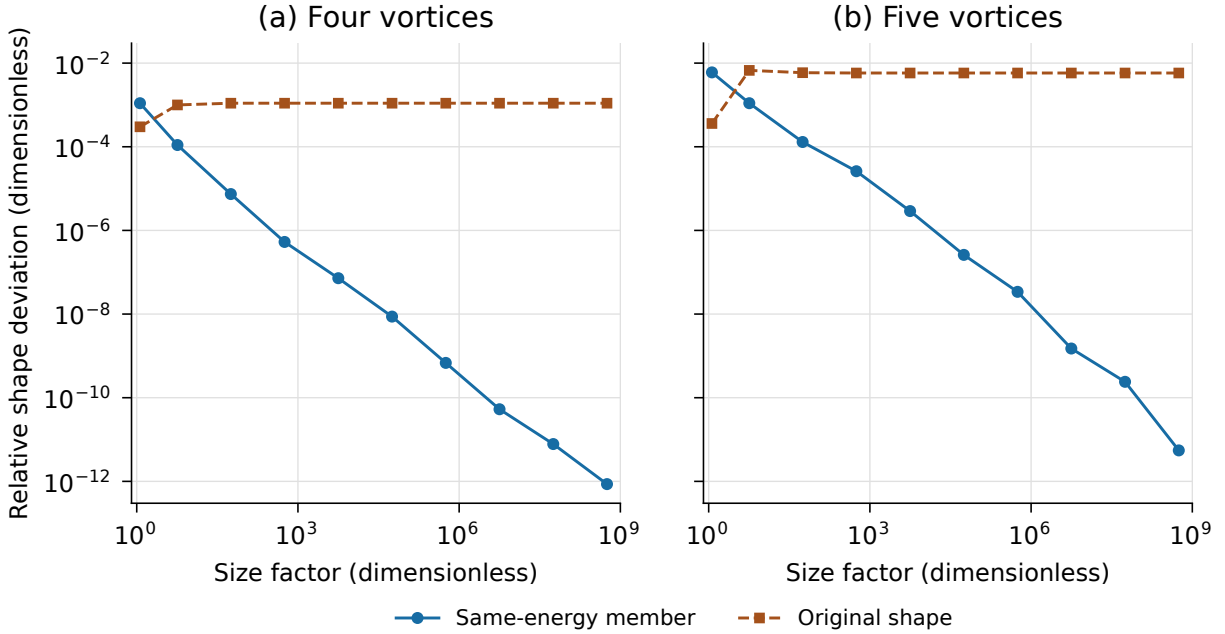


Figure 1: Recorded numerical illustrations of expansion after independent Gaussian perturbations of the real and imaginary coordinates with standard deviation 10^{-4} : (a) four vortices and (b) five vortices. The relative shape deviation is the Euclidean residual after removing translation and fitting a complex scale (rotation and dilation), divided by the norm of the translated configuration. The size factor is the modulus of that fitted scale relative to the original shape. Both axes are dimensionless. Dots are the ten rounded samples printed for each configuration in `data/verify-stable-expansion.txt`; lines only join those samples. The comparison with the family member having the perturbed energy decays, while the comparison with the original shape levels off. This diagnostic is weaker than the theorem’s estimate and is a binary64 illustration, not an interval proof. The figure is reproduced by `plot_stable_expansion.py`.

is conserved to $4 \cdot 10^{-14}$. The reversal of the unstable four-vortex control, perturbed by 10^{-8} in each coordinate, departs from its shape by $1.5 \cdot 10^{-3}$ when its size has grown eightfold and by 0.3 at sixteenfold; while the deviation lies between 10^{-7} and 10^{-2} it grows like the size to a power between 5.9 and 6.0, close to the value $5.981\dots$ of the certified $-k$, and the program checks that each measured power lies within 0.2 of it. The reversal of the five-vortex control, perturbed in the same way, departs by $2.3 \cdot 10^{-4}$ at eightfold and by $4 \cdot 10^{-3}$ at eighteenfold growth. A self-similar collapse, by contrast, departs from self-similarity in binary64 under round-off alone [13, 7].

Figure 1 displays the stored comparisons with the original shape and with the family member selected by energy.

How common. A least-squares solver from random starts, in binary64 with fixed seeds, gave 342 converged four-vortex collapses (not checked for duplicates), of which 52 have $c > 0$, and 543 converged five-vortex collapses (also not checked for duplicates), of which 32 have every remaining exponent with positive real part (`survey_expansions.py`). This is a sample of what a naive search finds, not a measure on the collapse set. Of the 52 four-vortex cases, 37 have the remaining pair on $\operatorname{Re} k = 1$, as in Theorem 1, and 15 a real pair $0 < k < 2$. Of the 32 five-vortex cases, 10 have both remaining pairs on $\operatorname{Re} k = 1$, as in Theorem 2, 16 a quadruple $1 \pm x \pm iy$ with $0 < x < 1$, and

program	what it checks	checks
<code>verify_stable_expansion.py</code>	Theorems 1, 2, the hypotheses of Lemma 1 and of Theorem 3 on the enclosures, three controls, controls of (5) and of the Krawczyk test, integrations	62 in ball or exact arithmetic, 8 in binary64
<code>survey_expansions.py</code>	the random sample of Section 6 (numerical)	none

Table 1: The programs. The verification program prints every check and, at the end (at once if the first Krawczyk test fails), exits with status 1 if any failed. Five of its checks in ball arithmetic are regression tests, identities that hold for every configuration ($\text{tr } DE = 2N$, $\text{Im tr}(A\bar{A}) = 0$, $\text{tr}((DE - I)^2) = 2\text{tr}(A\bar{A}) - 2Nb^2$ at two configurations, and Vieta’s formulas for the roots u_1, u_2 computed from m_1 and m_2): tests of how DE is assembled from A , of the trace formula and of the roots, not of the theorems, and the output labels them so. Two of its checks are in exact rational arithmetic. The output is in `data/`.

6 at least one real pair $0 < k < 2$.

7 Discussion

The theorems give self-similarly expanding configurations of four and five vortices that are linearly asymptotically stable modulo the rotation and the family of expansions with the same circulations, with a spectral gap. Theorem 3 makes this nonlinear: in similarity variables, nearby motions converge, up to a rotation drifting at the rate $b(\sigma_\infty) - b^*$ and at the rate the linearization predicts, to the member of the family with their own energy, and in the original variables they stay within a bounded distance of an exactly self-similar expansion. Zbarsky writes that orbital linear stability, “which may not be hard to obtain for some systems, is not enough” [17]. What his argument needs is an estimate for approximate solutions whose error is small and decays, with constants that do not depend on the length of time, and Proposition 1 is such an estimate. Theorem 4 is then his theorem for these two configurations. Its proof follows his argument, and Appendix A gives the estimates for N patches in full: the other patches act on a patch only through the strain of their field, and a single renormalizing function built from that strain replaces his functions for the two other patches. Small slips of the arXiv version are corrected there; none affects his conclusions.

The examples are Euler vortices. The α -models, where a vortex induces the velocity $\Gamma r^{-\alpha-1}/(2\pi)$, have the same structure after the obvious changes of homogeneity, and we have not treated them. Nor have we asked which fraction of the collapse set reverses into stable expansions, or whether stable expansions exist for every $N \geq 4$; the sample of Section 6 suggests that they are common for $N = 4$ and 5.

Data availability. The programs `verify_stable_expansion.py` and `survey_expansions.py`, with the modules `certify_ball_ad.py` and `certify_pipeline.py` of [8], their inputs and their output are in the repository <https://github.com/ChaseHendrick/stable-expansion>: the programs in `code/` and their inputs and output in `data/`. Release 1.0.0 of that repository, which holds the programs and data of this version of the paper, is archived at Zenodo, <https://doi.org/10.5281/zenodo.22971173>. Release 1.0.2 archives the same proof programs and data with supplementary checks at <https://doi.org/10.5281/zenodo.23047037>.

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A The estimates for the patches

This appendix proves the estimates used in steps (a) and (c) of the proof of Theorem 4, for any number $N \geq 3$ of patches (so many are needed for $\sum_{i < j} \Gamma_i \Gamma_j = 0$ with every $\Gamma_j \neq 0$) whose centres satisfy the bootstrap assumption (B1). The structure and the ideas are Zbarsky's [17]: the moments $I_{n,j}$, their renormalization, the concentration of each patch by the conservation of energy, and the averaging of the distance of a point of a patch to its centre over one turn. The patches other than ω_i enter only through the field they induce near ω_i (Lemma 3), and a single renormalizing function, built from the strain matrix of that field, replaces the functions $f_{k,2}$, $f_{k,3}$ of [17], one for each other patch. We work in the units of (1), $K(x) = x^\perp / (2\pi|x|^2)$, so that $u = K * \omega$.

A.1 Notation and consequences of the bootstrap

Put $\Omega_j = \int \omega_j = -\Gamma_j \neq 0$ and $m_j = \int |\omega_j| = |\Omega_j|$; each ω_j has the sign of Ω_j , so $\omega_j = \text{sgn}(\Omega_j)|\omega_j|$. Write r_{in} for the radius r of the initial supports in Theorem 4, since from now on r denotes a distance: for a fixed i write $\bar{x} = \bar{x}_i$, $r = r(x) = |x - \bar{x}_i|$, $I_n = I_{n,i} = \int r^n |\omega_i|$ and $m = m_i$. The bootstrap assumptions, with $\varepsilon \leq 1/18$, an even integer $n \geq 4$ and M as in the theorem, hold on $[t_0, T_*)$ as stated below and at T_* with $\leq$ in place of $<$: (B1) $\bar{x} = e^{i\beta}\gamma z$ with $z \in S_Y$, $|z - Y| < \varepsilon^2$ and $|\gamma/\sqrt{t} - 1| < \varepsilon$; (B2) $I_{2,j} < t^{\varepsilon/2}$; (B3) $I_{n,j} < t^{n(1+\varepsilon)/4}$; (B4) ω_j has the sign of Ω_j and $\|\omega_j\|_{L^{2+\varepsilon}} \leq M$; (B5) $\text{supp } \omega_j \subseteq B(\bar{x}_j, \varepsilon t^{1/4+\varepsilon})$. Below, C denotes constants that depend only on $Y, \Gamma, \varepsilon, M, r_{\text{in}}$ and n , possibly different at each occurrence, and never on χ, R, t_0 or T_* ; C_n records a dependence on n where it matters. All statements hold for $t_0 \leq t \leq T_*$ and $t_0 \geq T$, with T large enough, chosen last.

The solution is Yudovich's ([16]; for the whole plane [4, Theorems 1.6 and 3.3] and [1, Remark 2.4]): u is bounded, log-Lipschitz in x and continuous in (t, x) ; its flow $X(t, a)$, the position at time t of the particle that is at a at time t_0 , is for each t a measure-preserving homeomorphism of $\mathbb{R}^2$, and it is C^1 in t with $\partial_t X = u(t, X)$; and each $\omega_j(t)$ is $\omega_j(t_0)$ transported by X . For every $\phi \in C^1(\mathbb{R} \times \mathbb{R}^2)$

$$\frac{d}{dt} \int \phi(t, x) |\omega_j(t, x)| dx = \int (\partial_t \phi + u \cdot \nabla \phi) |\omega_j| dx, \quad (11)$$

and the same holds with ω_j in place of $|\omega_j|$, because the left side is the derivative of $\int \phi(t, X(t, a)) |\omega_j(t_0, a)| da$ and $\text{supp } \omega_j(t_0)$ is compact, so one may differentiate under the integral. With $\phi = x$, (11) gives $d\bar{x}_j/dt = \Omega_j^{-1} \int u \omega_j = \Omega_j^{-1} \int u(t, X(t, a)) \omega_j(t_0, a) da$, which is continuous in t because u is continuous in (t, x) . So each centre $\bar{x}_j$ is C^1 in t , and the test functions below, which depend on t through the centres, are admissible in (11). L^p norms and signs are preserved, so (B4) holds for all t . The other quantities in (B1)–(B5) are continuous in t : the moments by (11), those in (B1) because $\bar{x}$ is continuous, and the radius $\max_{p \in \text{supp } \omega_j(t)} |p - \bar{x}_j(t)|$ because $\text{supp } \omega_j(t) = X(t, \text{supp } \omega_j(t_0))$ and X is uniformly continuous on compact subsets of $[t_0, \infty) \times \mathbb{R}^2$. As (B1)–(B3) and (B5) hold at t_0 with strict inequalities (proof of Theorem 4), a first time T_* at which one of them fails satisfies $T_* > t_0$, and they hold on $[t_0, T_*)$ and, with $\leq$, at T_* .

Let $d_Y = \min_{i \neq j} |Y_i - Y_j|$, $D_Y = \max_j |Y_j|$ and $c_Y = d_Y/4$. By (B1), $|\bar{x}_i - \bar{x}_j| = \gamma|z_i - z_j|$ with $\gamma \in ((1 - \varepsilon)\sqrt{t}, (1 + \varepsilon)\sqrt{t})$ and $|z_j - Y_j| < \varepsilon^2$, so for ε small

$$2c_Y\sqrt{t} \leq |\bar{x}_i - \bar{x}_j| \leq 3D_Y\sqrt{t}, \quad |\bar{x}_j| \leq 2D_Y\sqrt{t}, \quad (12)$$

and for T large every $x \in B(\bar{x}_i, 2\varepsilon t^{1/4+\varepsilon})$ and every $y \in B(\bar{x}_j, \varepsilon t^{1/4+\varepsilon})$, $j \neq i$, satisfy $|x - y| \geq 2c_Y\sqrt{t} - 3\varepsilon t^{1/4+\varepsilon} \geq c_Y\sqrt{t}$; by (B5) the second ball contains $\text{supp } \omega_j$. We use $|K(z)| \leq C|z|^{-1}$, $|\nabla K(z)| \leq C|z|^{-2}$ and $|\nabla^2 K(z)| \leq C|z|^{-3}$.

Lemma 3 (The field of the other patches). *Let $V_i = \sum_{j \neq i} K * \omega_j$. For $x \in B(\bar{x}_i, 2\varepsilon t^{1/4+\varepsilon})$, $|V_i(x)| \leq Ct^{-1/2}$, $|\nabla V_i(x)| \leq Ct^{-1}$ and $|\nabla^2 V_i(x)| \leq Ct^{-3/2}$, and*

$$V_i(x) = V_i^0 + \Lambda_i(x - \bar{x}_i) + \Theta_i(x), \quad |\Theta_i(x)| \leq Ct^{-3/2}(|x - \bar{x}_i|^2 + \sum_j I_{2,j}),$$

with $V_i^0 = \sum_{j \neq i} \Omega_j K(\bar{x}_i - \bar{x}_j)$ and $\Lambda_i = \sum_{j \neq i} \Omega_j \nabla K(\bar{x}_i - \bar{x}_j)$. The matrix Λ_i is symmetric with trace zero, and $|\Lambda_i| \leq Ct^{-1}$.

Proof. The bounds on V_i and its derivatives follow from $|x - y| \geq c_Y \sqrt{t}$ on the supports. For $j \neq i$ and $y \in \text{supp } \omega_j$, the segment from $x - \bar{x}_j$ to $x - y$ consists of the points $x - y'$ with $y' \in [\bar{x}_j, y] \subseteq B(\bar{x}_j, \varepsilon t^{1/4+\varepsilon})$, so $|\nabla^2 K| \leq C(c_Y \sqrt{t})^{-3}$ on it, and Taylor's formula gives $K(x - y) = K(x - \bar{x}_j) + \nabla K(x - \bar{x}_j)(\bar{x}_j - y) + O(t^{-3/2}|y - \bar{x}_j|^2)$; integrating against ω_j , the linear term vanishes because $\bar{x}_j$ is the centre of vorticity, so $K * \omega_j(x) = \Omega_j K(x - \bar{x}_j) + O(t^{-3/2} I_{2,j})$. Expanding $K(x - \bar{x}_j)$ about $\bar{x}_i - \bar{x}_j$ in the same way, along the points $x' - \bar{x}_j$ with $x' \in [\bar{x}_i, x]$, gives the formula. With $G(z) = \log |z|/(2\pi)$ and J the rotation by $\pi/2$, $K = J\nabla G$, so $\nabla K = J\nabla^2 G$; $\nabla^2 G$ is symmetric with trace $\Delta G = 0$, and for $S = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ one has $JS = \begin{pmatrix} -b & a \\ a & b \end{pmatrix}$, again symmetric with trace zero. $\square$

Lemma 4 (The centres). $d\bar{x}_i/dt = V_i^0 + e_i(t)$ with $|e_i| \leq Ct^{-3/2} \sum_j I_{2,j} \leq Ct^{\varepsilon/2-3/2}$. Hence $|d\bar{x}_i/dt| \leq Ct^{-1/2}$ and $|d\Lambda_i/dt| \leq Ct^{-2}$.

Proof. By (11) with $\phi(x) = x$, $\Omega_i d\bar{x}_i/dt = \iint K(x - y) \omega_i(x) \omega_i(y) dx dy + \int V_i \omega_i$. The first integral converges absolutely, since $|K(z)| \leq 1/(2\pi|z|)$ and $\omega_i \in L^1 \cap L^{2+\varepsilon}$, and vanishes because K is odd. By Lemma 3 the second is $\Omega_i V_i^0 + \Lambda_i \int (x - \bar{x}_i) \omega_i + \int \Theta_i \omega_i = \Omega_i V_i^0 + O(t^{-3/2}(I_{2,i} + m_i \sum_j I_{2,j}))$. Finally $|V_i^0| \leq Ct^{-1/2}$, and $d\Lambda_i/dt = \sum_{j \neq i} \Omega_j \nabla^2 K(\bar{x}_i - \bar{x}_j)[d\bar{x}_i/dt - d\bar{x}_j/dt]$ with $|\nabla^2 K(\bar{x}_i - \bar{x}_j)| \leq Ct^{-3/2}$. $\square$

A.2 The moments

Let $q(x) = (x - \bar{x}) \cdot \Lambda_i(x - \bar{x})$, so that $|q| \leq Ct^{-1}r^2$. For vectors a, b write $[a, b] = a_1 b_2 - a_2 b_1$, so that $a \cdot b^\perp = -[a, b]$ and $[a, b] = [a, b - a]$.

Lemma 5 (The moments). *For every even $n \geq 2$,*

$$\frac{d}{dt} I_n = A_n + n \int r^{n-2} q |\omega_i| + B_n,$$

where $A_2 = 0$, $|A_n| \leq C_n I_2 I_n^{(n-4)/n}$ for $n \geq 4$, and $|B_n| \leq C n t^{-3/2} (\sup_{\text{supp } \omega_i} r \cdot I_n + \sum_j I_{2,j} \cdot I_n^{(n-1)/n})$.

Proof. Apply (11) to $\phi = |x - \bar{x}(t)|^n$, a polynomial in x : $dI_n/dt = n \int r^{n-2} (x - \bar{x}) \cdot (u(x) - d\bar{x}/dt) |\omega_i| dx$. Split $u = K * \omega_i + V_i$. The part of $K * \omega_i$ is $A_n = n \iint r_x^{n-2} (x - \bar{x}) \cdot K(x - y) |\omega_i(x) \omega_i(y)| dx dy$, with $r_x = r(x)$, $r_y = r(y)$. By Lemmas 3 and 4, $V_i(x) - d\bar{x}/dt = \Lambda_i(x - \bar{x}) + \Theta_i(x) - e_i$, which gives the middle term and $B_n = n \int r^{n-2} (x - \bar{x}) \cdot (\Theta_i - e_i) |\omega_i|$; the bound on B_n follows with $I_{n-1} \leq m^{1/n} I_n^{(n-1)/n}$ (Hölder's inequality).

For A_n , note $(x - \bar{x}) \cdot (x - y)^\perp = (x - \bar{x}) \cdot (\bar{x} - y)^\perp = [x - \bar{x}, y - \bar{x}]$. For $n = 2$ the integrand is $2 \text{sgn}(\Omega_i) [x - \bar{x}, y - \bar{x}] |x - y|^{-2} |\omega_i(x) \omega_i(y)| / 2\pi$, antisymmetric in x and y and absolutely integrable (it is at most $r_x |x - y|^{-1}$), so $A_2 = 0$. For $n \geq 4$, since $\int (\bar{x} - y)^\perp \omega_i(y) dy = 0$ we may subtract $r_x^{n-4} (x - \bar{x}) \cdot (\bar{x} - y)^\perp$ from the kernel, so

$$A_n = \frac{n}{2\pi} \iint \Phi(x, y) |\omega_i(x) \omega_i(y)| dx dy, \quad \Phi = r_x^{n-2} (x - \bar{x}) \cdot (\bar{x} - y)^\perp \left(\frac{1}{|x - y|^2} - \frac{1}{r_x^2} \right).$$

On $S_1 = \{r_x < r_y/2\}$, $|x - y| \geq r_y/2$ and $|\Phi| \leq r_x^{n-1} r_y (4r_y^{-2} + r_x^{-2}) \leq r_x^{n-4} r_y^2$. On $S_3 = \{r_x > 2r_y\}$, $|x - y| \geq r_x/2$ and $||x - y|^{-2} - r_x^{-2}| \leq (2r_x r_y + r_y^2) 4r_x^{-4} \leq 10r_y r_x^{-3}$, so $|\Phi| \leq 10r_x^{n-4} r_y^2$. On $S_2 = \{r_y/2 \leq r_x \leq 2r_y\}$ the subtracted term is at most $r_x^{n-3} r_y \leq 2r_x^{n-4} r_y^2$; the other term, $r_x^{n-2} [x - \bar{x}, y - \bar{y}] |x - y|^{-2}$, has an antisymmetric numerator and S_2 is symmetric, so its integral equals half the integral of $(r_x^{n-2} - r_y^{n-2}) [x - \bar{x}, y - \bar{y}] |x - y|^{-2}$. Here $r_x^{n-2} - r_y^{n-2} = (r_x^2 - r_y^2) P$ with $0 \leq P \leq \frac{n-2}{2} \max(r_x, r_y)^{n-4}$, $|r_x^2 - r_y^2| = |(x - y) \cdot (x + y - 2\bar{x})| \leq |x - y| (r_x + r_y)$ and $|[x - \bar{x}, y - \bar{y}]| = |[x - \bar{x}, y - x]| \leq r_x |x - y|$, so this term is at most $C_n r_x^{n-4} r_y^2$ on S_2 . Altogether $|A_n| \leq C_n \iint r_x^{n-4} r_y^2 |\omega_i(x)| |\omega_i(y)| = C_n I_{n-4} I_2 \leq C_n m^{4/n} I_n^{(n-4)/n} I_2$. $\square$

A.3 Concentration and renormalization

Lemma 6 (Conservation of the pseudo-energy). *For every $t \geq t_0$ the integral*

$$\mathcal{E}(t) = \iint \log |x - y| \omega(t, x) \omega(t, y) dx dy$$

converges absolutely, and $\mathcal{E}(t) = \mathcal{E}(t_0)$.

Proof. We use only the properties of the solution stated before (11); the log-Lipschitz bound enters only through the existence of the flow. Fix $T > t_0$, write $\omega_0 = \omega(t_0)$, $L(z) = \log |z|$, so that $\nabla L(z) = z/|z|^2$, and $U = \sup |u|$, and choose $R \geq 1$ with $\text{supp } \omega_0 \subseteq B(0, R - U(T - t_0))$. For $t \in [t_0, T]$, $\omega(t) = \omega_0 \circ X(t, \cdot)^{-1}$ with $X(t, \cdot)$ a measure-preserving homeomorphism, so $\|\omega(t)\|_{L^1} = \|\omega_0\|_{L^1}$ and $\|\omega(t)\|_{L^\infty} = \|\omega_0\|_{L^\infty}$, and $\text{supp } \omega(t) \subseteq X(t, \text{supp } \omega_0) \subseteq B(0, R)$ because $|\partial_t X| \leq U$. Splitting the integrals at $|x - y| = 1$ gives, for $x \in B(0, R)$ in the first bound and every x in the second,

$$\int |L(x - y)| |\omega(t, y)| dy \leq \frac{\pi}{2} \|\omega_0\|_{L^\infty} + \log(2R) \|\omega_0\|_{L^1}, \quad \int \frac{|\omega(t, y)|}{|x - y|} dy \leq 2\pi \|\omega_0\|_{L^\infty} + \|\omega_0\|_{L^1}. \quad (13)$$

So $\mathcal{E}(t)$ converges absolutely, and so does

$$D(t) = \iint \nabla L(x - y) \cdot (u(t, x) - u(t, y)) \omega(t, x) \omega(t, y) dx dy,$$

whose integrand is at most $2U|x - y|^{-1} |\omega(t, x)| |\omega(t, y)|$.

In these two integrals change the variables $x = X(t, a)$ and $y = X(t, b)$, one at a time (Fubini's theorem). With

$$F(t, a, b) = \nabla L(X(t, a) - X(t, b)) \cdot (u(t, X(t, a)) - u(t, X(t, b))),$$

this gives $\mathcal{E}(t) = \iint L(X(t, a) - X(t, b)) \omega_0(a) \omega_0(b) da db$ and $D(t) = \iint F(t, a, b) \omega_0(a) \omega_0(b) da db$, and, by (13), $\iint |F(t, a, b)| \omega_0(a) |\omega_0(b)| da db \leq 2U \|\omega_0\|_{L^1} (2\pi \|\omega_0\|_{L^\infty} + \|\omega_0\|_{L^1})$ for every $t \in [t_0, T]$. For $a \neq b$, $X(t, a) \neq X(t, b)$ because $X(t, \cdot)$ is injective, so $t \mapsto L(X(t, a) - X(t, b))$ is C^1 on $[t_0, T]$ with derivative $F(\cdot, a, b)$, since $\partial_t X = u(t, X)$. As u is continuous and so is X (each $X(t, \cdot)$ is, and $|\partial_t X| \leq U$), F is continuous off the null set $\{a = b\}$, and by the bound just proved it is integrable on $[t_0, t] \times \mathbb{R}^2 \times \mathbb{R}^2$ for the measure $ds |\omega_0(a)| |\omega_0(b)| da db$. Fubini's theorem therefore gives

$$\mathcal{E}(t) - \mathcal{E}(t_0) = \iint \left(\int_{t_0}^t F(s, a, b) ds \right) \omega_0(a) \omega_0(b) da db = \int_{t_0}^t D(s) ds.$$

It remains to show that $D(s) = 0$. Fix s and write $\omega = \omega(s)$ and $u = u(s)$. The parts of $D(s)$ with $u(x)$ and with $u(y)$ converge absolutely by themselves, and exchanging the names of x and

y , with $\nabla L(y - x) = -\nabla L(x - y)$, turns $\iint \nabla L(x - y) \cdot u(y) \omega(x) \omega(y) dx dy$ into $-\iint \nabla L(x - y) \cdot u(x) \omega(x) \omega(y) dx dy$. Hence, by Fubini's theorem,

$$D(s) = 2 \int \omega(x) u(x) \cdot w(x) dx, \quad w(x) = \int \nabla L(x - y) \omega(y) dy,$$

where the integral defining $w(x)$ converges absolutely by (13). Since $K(z) = z^\perp / (2\pi|z|^2) = \nabla L(z)^\perp / 2\pi$, $u(x) = (K * \omega)(x) = w(x)^\perp / 2\pi$, which is orthogonal to $w(x)$. So $D(s) = 0$, and $\mathcal{E}(t) = \mathcal{E}(t_0)$ for $t \in [t_0, T]$; T was arbitrary. $\square$

Lemma 7 (Concentration). *There is a constant C_E such that $\iint |\omega_i(x)| |\omega_i(y)| \log^+ |x - y| dx dy \leq C_E$ on $[t_0, T_*]$ for every i . Hence for every $\chi \in (0, 1)$ there is $R = R(\chi) \geq 1$, depending on χ and C , such that at every t each patch has a point $\tilde{x}_i(t)$ with $\int_{|x - \tilde{x}_i| > R} |\omega_i| dx \leq \chi^4 m_i$.*

Proof. By Lemma 6, the pseudo-energy $\mathcal{E} = \iint \log |x - y| \omega(x) \omega(y) dx dy$ takes the same value at every $t \in [t_0, T_*]$. Write $\mathcal{E} = \sum_i \mathcal{E}_{ii} + \sum_{i \neq j} \mathcal{E}_{ij}$ with $\mathcal{E}_{ij} = \iint \log |x - y| \omega_i(x) \omega_j(y)$. For $i \neq j$, $x \in \text{supp } \omega_i$ and $y \in \text{supp } \omega_j$, by (B5) and (12) $\log |x - y| = \log |\bar{x}_i - \bar{x}_j| + O(t^{-1/4+\varepsilon})$, and by (B1) $\log |\bar{x}_i - \bar{x}_j| = \frac{1}{2} \log t + \log |Y_i - Y_j| + O(\varepsilon)$. So $\sum_{i \neq j} \mathcal{E}_{ij} = \frac{1}{2} \log t \sum_{i \neq j} \Omega_i \Omega_j + O(1) = O(1)$, because $\sum_{i \neq j} \Omega_i \Omega_j = 2 \sum_{i < j} \Gamma_i \Gamma_j = 0$, uniformly in $t \in [t_0, T_*]$. At t_0 also $\mathcal{E}_{ii} = O(1)$, since the supports have radius at most $2r_{\text{in}}$ and the negative part is bounded as below. By conservation, $\sum_i \mathcal{E}_{ii}(t) = O(1)$. Now $\mathcal{E}_{ii} = P_i - Q_i$ with $P_i = \iint |\omega_i| |\omega_i| \log^+ |x - y| \geq 0$ and, by Hölder's inequality in y with exponents $2 + \varepsilon$ and $(2 + \varepsilon)/(1 + \varepsilon)$,

$$0 \leq Q_i = \iint |\omega_i(x)| |\omega_i(y)| \log^- |x - y| \leq m_i \|\omega_i\|_{L^{2+\varepsilon}} \|\log^- |\cdot|\|_{L^{(2+\varepsilon)/(1+\varepsilon)}(B(0,1))} \leq C.$$

Hence $\sum_i P_i \leq C$ and each $P_i \leq C_E$. Take $R = \max(e, \exp(C_E/(\chi^4 \min_j m_j^2)))$; then $\iint_{|x-y|>R} |\omega_i(x)| |\omega_i(y)| \leq C_E / \log R \leq \chi^4 m_i^2$, so the average over y , with weight $|\omega_i(y)|$, of $\int_{|x-y|>R} |\omega_i(x)| dx$ is at most $\chi^4 m_i$, and some $y = \tilde{x}_i$ in the support does not exceed the average. $\square$

Lemma 8 (Renormalization). *Let $F(t, x) = |x - \bar{x}|^n (x - \bar{x}) \cdot J \Lambda_i(x - \bar{x})$, and let $\chi \in (0, 1)$ with $\chi^{3n-5} \leq 16^{-n}$. For $t \geq t_\chi$,*

$$\frac{d}{dt} \int F |\omega_i| = \frac{\Omega_i}{\pi} \int r^{n-2} q |\omega_i| + O(t^{-1}(\chi I_n + \chi^{-n} R(\chi)^n)),$$

where the constant in O does not depend on χ .

Proof. F is a polynomial in x of degree $n + 2$ with $|F| \leq Ct^{-1}r^{n+2}$, $|\nabla F| \leq C_n t^{-1}r^{n+1}$ and $|\nabla^2 F| \leq C_n t^{-1}r^n$, since $|\Lambda_i| \leq Ct^{-1}$. In polar coordinates about $\bar{x}$, with $\Lambda_i = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$, one has $q = r^2(a \cos 2\theta + b \sin 2\theta)$ and $(x - \bar{x}) \cdot J \Lambda_i(x - \bar{x}) = r^2(a \sin 2\theta - b \cos 2\theta)$, so $\partial_\theta F = 2r^n q$ and

$$\nabla F(x) \cdot K(x - \bar{x}) = \frac{\partial_\theta F}{2\pi r^2} = \frac{r^{n-2} q}{\pi}. \quad (14)$$

By (11), $\frac{d}{dt} \int F |\omega_i| = \int \partial_t F |\omega_i| + \int \nabla F \cdot V_i |\omega_i| + \iint \nabla F(x) \cdot K(x - y) |\omega_i(x)| |\omega_i(y)|$. By Lemma 4, $|\partial_t F| \leq C_n t^{-1} r^{n+1} |d\bar{x}/dt| + r^{n+2} |d\Lambda_i/dt| \leq C_n t^{-1} (t^{-1/2} r^{n+1} + t^{-1} r^{n+2})$, and $|V_i| \leq Ct^{-1/2}$; with $r \leq \varepsilon t^{1/4+\varepsilon}$ on the support, the first two integrals are $O(t^{-1} \cdot t^{-1/4+\varepsilon} I_n)$, which is at most $t^{-1} \chi I_n$ for $t \geq t_\chi$. By (14) and $\int \omega_i = \Omega_i$, the double integral is

$$\frac{\Omega_i}{\pi} \int r^{n-2} q |\omega_i| + \iint \nabla F(x) \cdot (K(x - y) - K(x - \bar{x})) |\omega_i(x)| |\omega_i(y)| dx dy.$$

We bound the last integral on $S_4 = \{r_x \leq \chi r_y\}$, $S_5 = \{\chi r_y < r_x < r_y/\chi\}$ and $S_6 = \{r_x \geq r_y/\chi\}$. On S_6 , $|y - \bar{x}| \leq \chi r_x$, so $|K(x - y) - K(x - \bar{x})| \leq C r_y / r_x^2 \leq C \chi / r_x$ and the integrand is at most $C_n \chi t^{-1} r_x^n$. On S_4 , $|x - y| \geq r_y/2$, so $|\nabla F(x) \cdot K(x - y)| \leq C_n t^{-1} r_x^{n+1} / r_y \leq C_n t^{-1} r_x^n$, and $|\nabla F(x) \cdot K(x - \bar{x})| \leq C_n t^{-1} r_x^n$ on $S_4 \cup S_5$. On S_5 , which is symmetric in x and y , the integral of $\nabla F(x) \cdot K(x - y)$ against the symmetric weight $\text{sgn}(\Omega_i) |\omega_i(x)| |\omega_i(y)|$ equals half the integral of $(\nabla F(x) - \nabla F(y)) \cdot K(x - y)$, and $|\nabla F(x) - \nabla F(y)| \leq C_n t^{-1} |x - y| (r_x^n + r_y^n)$, because every point of the segment $[x, y]$ lies within $\max(r_x, r_y)$ of $\bar{x}$; by symmetry again the r_y^n part has the same integral as the r_x^n part. Altogether the last integral is at most $C_n t^{-1} (\chi m I_n + \mathcal{S})$ with $\mathcal{S} = \iint_{S_4 \cup S_5} r_x^n |\omega_i(x)| |\omega_i(y)|$.

It remains to show $\mathcal{S} \leq 2\chi m I_n + m^2 (3R/\chi)^n$, with $\tilde{x} = \tilde{x}_i$ and R from Lemma 7. On $S_4 \cup S_5$, $r_x < r_y/\chi$. If $a = |\tilde{x} - \bar{x}| < 2R$: where $r_y \geq 3R$, $|y - \tilde{x}| > R$, so the y -mass is at most $\chi^4 m$ and this part is at most $\chi^4 m I_n$; where $r_y < 3R$, $r_x < 3R/\chi$ and this part is at most $m^2 (3R/\chi)^n$. If $a \geq 2R$: on $B(\tilde{x}, R)$, $(x - \bar{x}) \cdot (\tilde{x} - \bar{x})/a \geq a/2$ and the mass is at least $(1 - \chi^4)m$, so $\int_{B(\tilde{x}, R)} (x - \bar{x}) |\omega_i|$ has length at least $ma/4$; as $\int (x - \bar{x}) |\omega_i| = 0$, $\int_{\mathbb{R}^2 \setminus B(\tilde{x}, R)} r |\omega_i| \geq ma/4$, and the part of it where $r \leq \varrho_0 = a/(8\chi^4)$ is at most $\varrho_0 \chi^4 m = ma/8$. By Hölder's inequality on $\{r > \varrho_0\} \setminus B(\tilde{x}, R)$, whose mass is at most $\chi^4 m$, $ma/8 \leq I_n^{1/n} (\chi^4 m)^{(n-1)/n}$, that is $I_n \geq ma^n 8^{-n} \chi^{-4(n-1)}$. Then the part of $\mathcal{S}$ where $r_x < 2a/\chi$ is at most $m^2 (2a/\chi)^n \leq m 16^n \chi^{3n-4} I_n \leq \chi m I_n$, and where $r_x \geq 2a/\chi$ we have $r_y > \chi r_x \geq 2a$, so $|y - \tilde{x}| > a \geq R$ and that part is at most $\chi^4 m I_n$. $\square$

Combining Lemmas 5 and 8: for $t \geq t_\chi$,

$$\frac{d}{dt} I_n = A_n + \frac{n\pi}{\Omega_i} \frac{d}{dt} \int F |\omega_i| + B_n + O(t^{-1} (\chi I_n + \chi^{-n} R(\chi)^n)), \quad \left| \int F |\omega_i| \right| \leq C \varepsilon^2 t^{-1/2+2\varepsilon} I_n. \quad (15)$$

Compared with [17, (12)–(14)], the dependence of F on t , through $\bar{x}$ and Λ_i , is accounted for by the term $\int \partial_t F |\omega_i|$ above, which the arXiv version omits from (34); the bound on Q_i uses the exponents $2 + \varepsilon$ and $(2 + \varepsilon)/(1 + \varepsilon)$ where the arXiv version writes an exponent below 1. Neither changes the conclusions of [17].

A.4 Long times

Proposition 2. *Let χ be small enough, depending on ε , n and C ; in particular χ satisfies the hypothesis $\chi^{3n-5} \leq 16^{-n}$ of Lemma 8 and the same hypothesis with 2 in place of n , which reads $\chi \leq 1/256$, since the proof uses that lemma for both orders. Let T be large enough, depending also on χ . If $T_* \geq t_0 + t_0^{9/10}$ and (B1)–(B5) hold on $[t_0, T_*)$, and so with $\leq$ at T_* , then at T_* , for every i , $I_{2,i} < T_*^{\varepsilon/2}$, $I_{n,i} < T_*^{n(1+\varepsilon)/4}$ and $\text{supp } \omega_i \subseteq B(\bar{x}_i, \varepsilon T_*^{1/4+\varepsilon})$ with room to spare: every point of the support is at distance less than $\varepsilon T_*^{1/4+\varepsilon}$ from $\bar{x}_i$.*

Proof. Let $t_1 = T_* - T_*^{2/3}$; then $t_1 \geq t_0$ and $T_*/2 \leq t \leq T_*$ on $[t_1, T_*]$. Integrate (15), for the orders 2 and n , over $[t_1, T_*]$ and use its second part at both ends. At T_* the assumptions are used only with $\leq$; the strict inequalities are used only at $t_1 < T_*$.

$n = 2$. $A_2 = 0$ and, by (B2), (B5) and $I_1 \leq m^{1/2} I_2^{1/2}$, $|B_2| \leq C t^{-5/4+2\varepsilon}$. Hence

$$I_2(T_*) \leq I_2(t_1) + C T_*^{-1/2+5\varepsilon/2} + C T_*^{2/3} (T_*^{-5/4+2\varepsilon} + \chi T_*^{\varepsilon/2-1} + \chi^{-2} R^2 T_*^{-1}).$$

Since $(1 - x)^{\varepsilon/2} \leq 1 - \varepsilon x/2$ for $0 \leq x \leq 1$, $I_2(t_1) < t_1^{\varepsilon/2} \leq T_*^{\varepsilon/2} - \frac{\varepsilon}{2} T_*^{\varepsilon/2-1/3}$, and every other term is $o(T_*^{\varepsilon/2-1/3})$ except $C \chi T_*^{\varepsilon/2-1/3}$. So $I_2(T_*) < T_*^{\varepsilon/2}$ once $C \chi \leq \varepsilon/4$ and T is large.

$n \geq 4$. Put $\ell = n(1+\varepsilon)/4 > 1$. By (B2), (B3) and (B5), $|A_n| \leq C_n t^{\ell-1-\varepsilon/2}$ and $|B_n| \leq C t^{\ell-5/4+\varepsilon}$, so

$$I_n(T_*) \leq I_n(t_1) + C\varepsilon^2 T_*^{\ell-1/2+2\varepsilon} + C_n T_*^{\ell-1/3-\varepsilon/2} + C T_*^{\ell-7/12+\varepsilon} + C\chi T_*^{\ell-1/3} + C\chi^{-n} R^n T_*^{-1/3},$$

and $I_n(t_1) < t_1^\ell \leq T_*^\ell - T_*^{\ell-1/3}$ because $(1-x)^\ell \leq 1-x$. As $\varepsilon < 1/12$, $I_n(T_*) < T_*^\ell$ once $C\chi \leq 1/2$ and T is large.

The support. Let p_* be a point of $\text{supp } \omega_i(T_*)$ and $p(t) = X(t; T_*, p_*)$, so that $p(t) \in \text{supp } \omega_i(t)$; let $h = p - \bar{x}$ and $\varrho = |h|$. By (B5), $\varrho(t) \leq \varepsilon t^{1/4+\varepsilon} \leq 2\varrho_*$ on $[t_1, T_*]$, where $\varrho_* = \frac{\varepsilon}{2} T_*^{1/4+\varepsilon}$. Let $t_2 = \sup\{s \in [t_1, T_*] : s = t_1 \text{ or } \varrho(s) \leq \varrho_*\}$. By continuity $\varrho(t_2) \leq \varrho_*$ if $t_2 > t_1$, and $\varrho_* \leq \varrho \leq 2\varrho_*$ on $[t_2, T_*]$ if $t_2 < T_*$. If $t_2 = T_*$, then $\varrho(T_*) \leq \varrho_*$ and there is nothing to prove, so let $t_2 < T_*$. On $[t_2, T_*]$, $\dot{h} = u(p) - d\bar{x}/dt = \Omega_i K(h) + \Lambda_i h + \eta$ with

$$\eta = w + \Theta_i(p) - e_i, \quad w = \int (K(p-x) - K(h))\omega_i(x) dx.$$

We claim $|w| \leq C T_*^{-3/4-5\varepsilon/2}$ once n is large enough, depending on ε . Let $S_7 = \{r_x < \varrho/2\}$ and $S_8 = \{r_x \geq \varrho/2\}$, and $N_h = \nabla K(h)$. On S_7 the segment from h to $p-x = h - (x-\bar{x})$ stays at distance at least $\varrho/2$ from 0, so $K(p-x) = K(h) - N_h(x-\bar{x}) + O(r_x^2 \varrho^{-3})$, and $\int_{S_7} N_h(x-\bar{x})\omega_i = -\int_{S_8} N_h(x-\bar{x})\omega_i$ because $\bar{x}$ is the centre. So $|w| \leq C\varrho^{-3}I_2 + \int_{S_8} (|K(p-x)| + |K(h)| + C\varrho^{-2}r_x)|\omega_i|$. By (B3), the mass of S_8 is at most $I_n(\varrho/2)^{-n} \leq (4/\varepsilon)^n T_*^{-3n\varepsilon/4}$, and $\int_{S_8} r_x |\omega_i| \leq I_n(\varrho/2)^{1-n}$. Fix $s = 2 + \varepsilon/2$ and $\varpi \in (0, 1)$ with $1/s = \varpi/(2+\varepsilon) + 1-\varpi$; by Hölder's inequality and interpolation,

$$\int_{S_8} \frac{|\omega_i(x)|}{|p-x|} dx \leq \| |\cdot|^{-1} \|_{L^{s/(s-1)}(B(0,1))} M^\varpi \left(\int_{S_8} |\omega_i| \right)^{1-\varpi} + \int_{S_8} |\omega_i|.$$

With $\varrho \geq \varrho_*$ and $I_2 \leq T_*^{\varepsilon/2}$ the term $C\varrho^{-3}I_2$ in the bound on $|w|$ is at most $C T_*^{-3/4-5\varepsilon/2}$, and the integral over S_8 is at most $C T_*^{-3n\varepsilon(1-\varpi)/4}$, which is smaller once $3n\varepsilon(1-\varpi)/4 \geq 3/4 + 5\varepsilon/2$. Here $1-\varpi = (\varepsilon/2)/((2+\varepsilon/2)(1+\varepsilon))$, which at $\varepsilon = 1/18$ is $18/1387$, so that the condition asks for $n \geq 1644$; this is harmless, since n is fixed before χ and T are chosen. By Lemmas 3 and 4, $|\Theta_i(p)| + |e_i| \leq C\varepsilon^2 T_*^{-1+2\varepsilon} \leq C T_*^{-3/4-5\varepsilon/2}$ as $\varepsilon \leq 1/18$. So $|\eta| \leq C T_*^{-3/4-5\varepsilon/2}$.

Since $h \cdot K(h) = 0$ and $h^\perp \cdot K(h) = 1/2\pi$,

$$\frac{d\varrho^2}{dt} = 2q(p) + O(\varrho|\eta|), \quad \dot{\varphi} = \frac{\Omega_i}{2\pi\varrho^2}(1 + \psi(t)), \quad |\psi| \leq C T_*^{-1/2+2\varepsilon},$$

where φ is a continuous argument of h , because $|h^\perp \cdot (\Lambda_i h + \eta)|/\varrho^2 \leq C(T_*^{-1} + |\eta|/\varrho)$. So φ is strictly monotone, and it turns by 2π in time at most $\tau_1 = C T_*^{1/2+2\varepsilon}$; since $|\dot{\varphi}| \leq |\Omega_i|(1+|\psi|)/(2\pi\varrho^2)$ and $\varrho_*^2 = \frac{\varepsilon^2}{4} T_*^{1/2+2\varepsilon}$, each turn by 2π also takes time at least $T_*^{1/2+2\varepsilon}/C$. Divide $[t_2, T_*]$, of length at most $T_*^{2/3}$, into consecutive full turns, at most $C T_*^{2/3}/T_*^{1/2+2\varepsilon}$ of them, and a last interval shorter than τ_1 . On a full turn, change the variable to φ : with $q = \varrho^2(a \cos 2\varphi + b \sin 2\varphi)$,

$$\int q dt = \frac{2\pi}{\Omega_i} \int \varrho^4 (a \cos 2\varphi + b \sin 2\varphi) (1 + \psi)^{-1} d\varphi.$$

Over the turn, ϱ^2 changes by at most $\tau_1 \cdot C(T_*^{-1}\varrho^2 + \varrho|\eta|) \leq C T_*^{4\varepsilon}$ and a, b by at most $\tau_1 |d\Lambda_i/dt| \leq C T_*^{-3/2+2\varepsilon}$, while $\varrho^4 |\Lambda_i| \leq C T_*^{4\varepsilon}$. Replacing $\varrho^4 a$, $\varrho^4 b$ and ψ by their values at the start of the turn changes the integral by $O(T_*^{6\varepsilon-1/2})$, and with frozen coefficients it vanishes. So the full turns contribute at most $C T_*^{2/3-1/2-2\varepsilon} T_*^{6\varepsilon-1/2} = C T_*^{4\varepsilon-1/3}$ to $\varrho^2(T_*) - \varrho^2(t_2)$, the last interval at most

$\tau_1 Ct^{-1}(2\varrho_*)^2 \leq CT_*^{4\varepsilon}$, and the term $O(\varrho|\eta|)$ at most $CT_*^{2/3}\varrho_*T_*^{-3/4-5\varepsilon/2} \leq CT_*^{1/6-3\varepsilon/2}$. If $t_2 > t_1$, then $\varrho(t_2) \leq \varrho_*$ and $\varrho(T_*)^2 \leq \varrho_*^2 + o(T_*^{1/2+2\varepsilon}) < 4\varrho_*^2$. If $t_2 = t_1$, then $\varrho(t_1)^2 \leq \varepsilon^2 t_1^{1/2+2\varepsilon} \leq \varepsilon^2 T_*^{1/2+2\varepsilon} - \frac{\varepsilon^2}{2} T_*^{1/6+2\varepsilon}$, and the three contributions are $o(T_*^{1/6+2\varepsilon})$ because $\varepsilon < 1/12$; so again $\varrho(T_*) < \varepsilon T_*^{1/4+\varepsilon}$. $\square$

A.5 Short times

Proposition 3. *With the same choices, if $T_* < t_0 + t_0^{9/10}$ and (B1)–(B5) hold on $[t_0, T_*)$, and so with $\leq$ at T_* , the conclusion of Proposition 2 holds at T_* .*

Proof. Here $t_0 \leq t \leq T_* \leq 2t_0$. At t_0 the support of ω_i lies within $2r_{\text{in}}$ of $\bar{x}_i(t_0)$, so $I_n(t_0) \leq (2r_{\text{in}})^n m$. By Lemma 5 and $|q| \leq Ct^{-1}r^2$, $dI_n/dt \leq Cnt^{-1}I_n + |A_n| + |B_n|$. For $n = 2$ this gives $dI_2/dt \leq Ct^{-1}I_2 + Ct^{-5/4+2\varepsilon}$, and since $\int_{t_0}^{T_*} Ct^{-1} dt \leq Ct_0^{-1/10}$, Gronwall's inequality gives $I_2(T_*) \leq 2(4r_{\text{in}}^2 m + 1) < T_*^{\varepsilon/2}$ for T large. For $n \geq 4$, (B2), (B3) and (B5) give $dI_n/dt \leq Cnt^{\ell-1}$, so $I_n(T_*) \leq (2r_{\text{in}})^n m + Cnt_0^{9/10} t_0^{\ell-1} < T_*^\ell$. For the support, with p, h, ϱ and ϱ_* as before and $\varrho(t_0) \leq 2r_{\text{in}} < \varrho_*$, let $t_2 = \sup\{s \in [t_0, T_*] : \varrho(s) \leq \varrho_*\} > t_0$, so that $\varrho(t_2) \leq \varrho_*$. If $t_2 = T_*$ there is nothing to prove. Otherwise $\varrho \geq \varrho_*$ on $[t_2, T_*]$, the bound $|\eta| \leq CT_*^{-3/4-5\varepsilon/2}$ holds there as before, and $|\Lambda_i h| \leq CT_*^{-1}\varrho_*$, so $\varrho(T_*) \leq \varrho_* + T_*^{9/10}C(T_*^{-3/4+\varepsilon}) < 2\varrho_* = \varepsilon T_*^{1/4+\varepsilon}$ for T large. $\square$

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